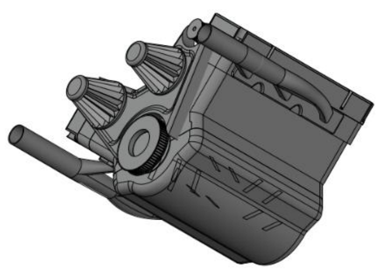
V-LINE 6 CYLINDER ENGINE

ENGINEERING DRAWING AND GRAPHICS LAB PROJECT

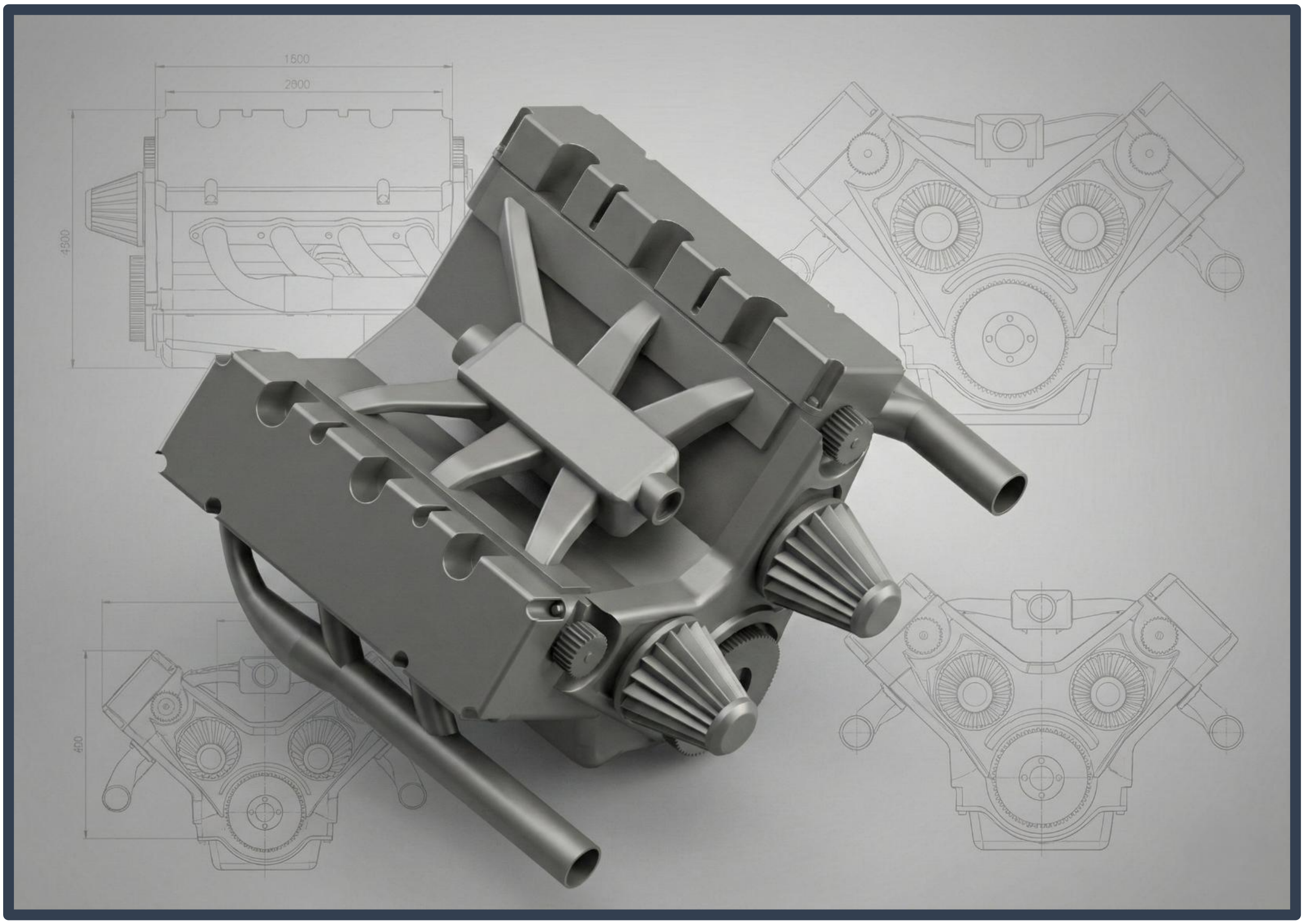
Project

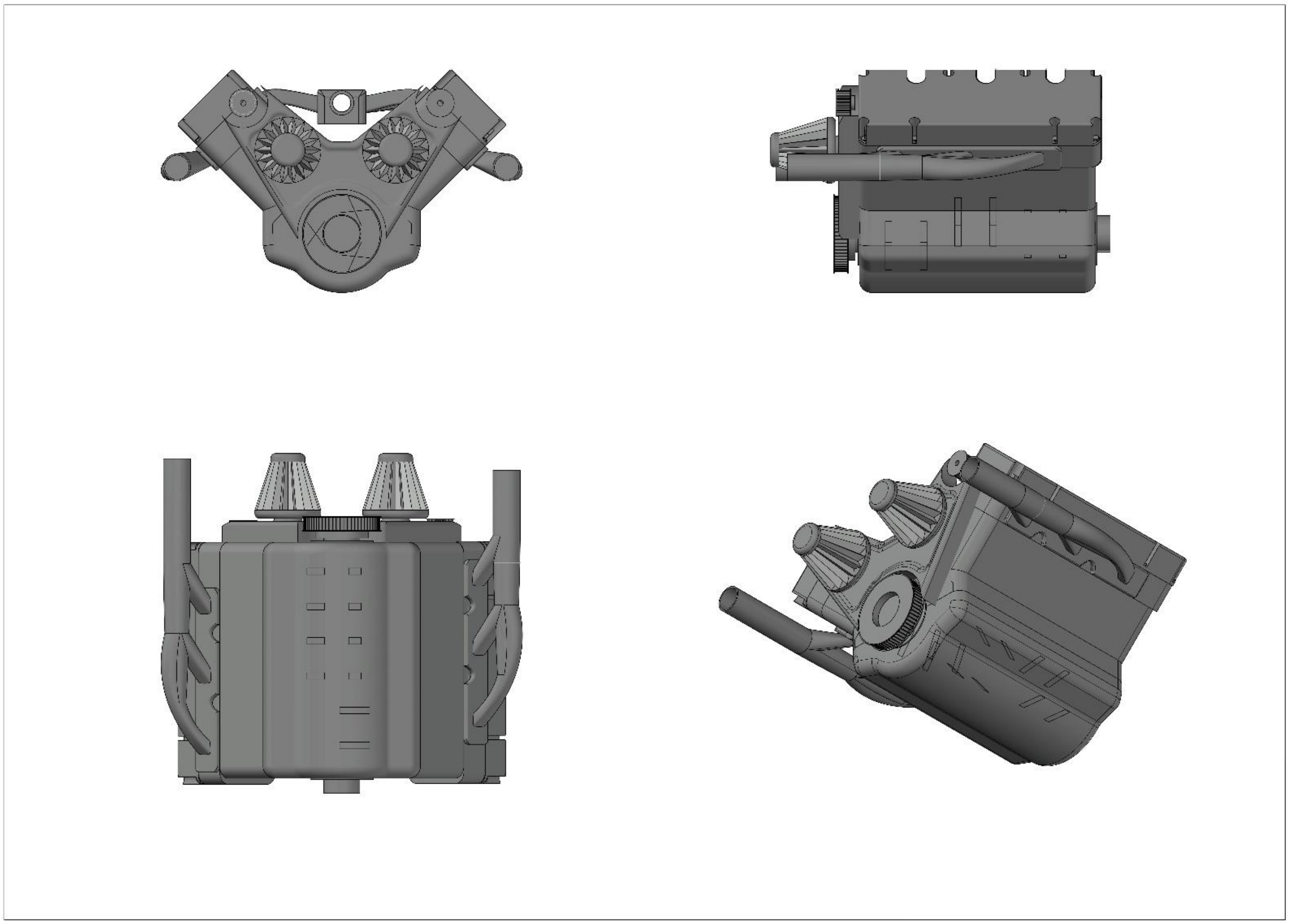


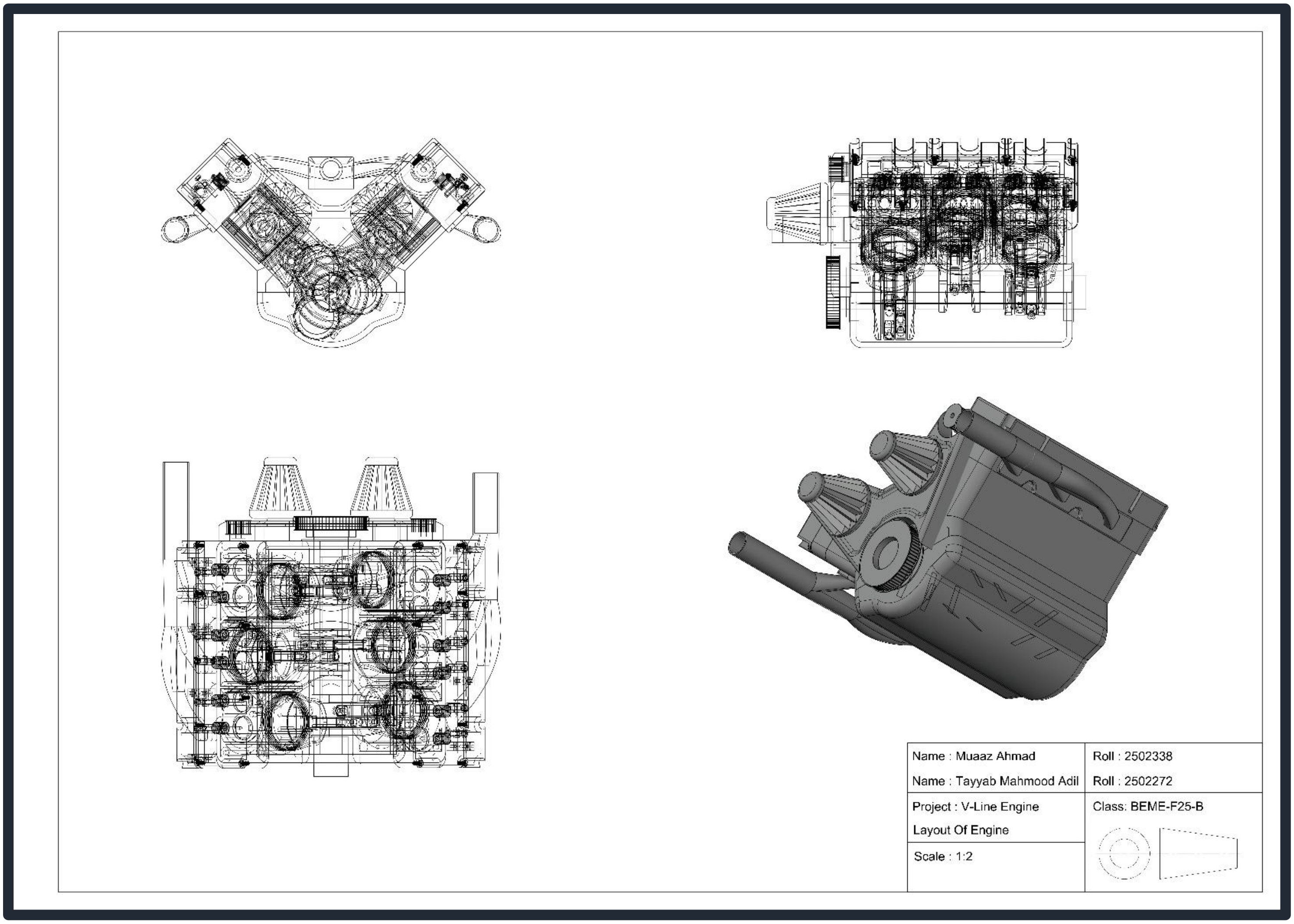
GROUP MEMBERS	
Muaaz Ahmad	2502338-B
Tayyab Mahmood Adil	2502272-B

DECEMBER 25, 2025

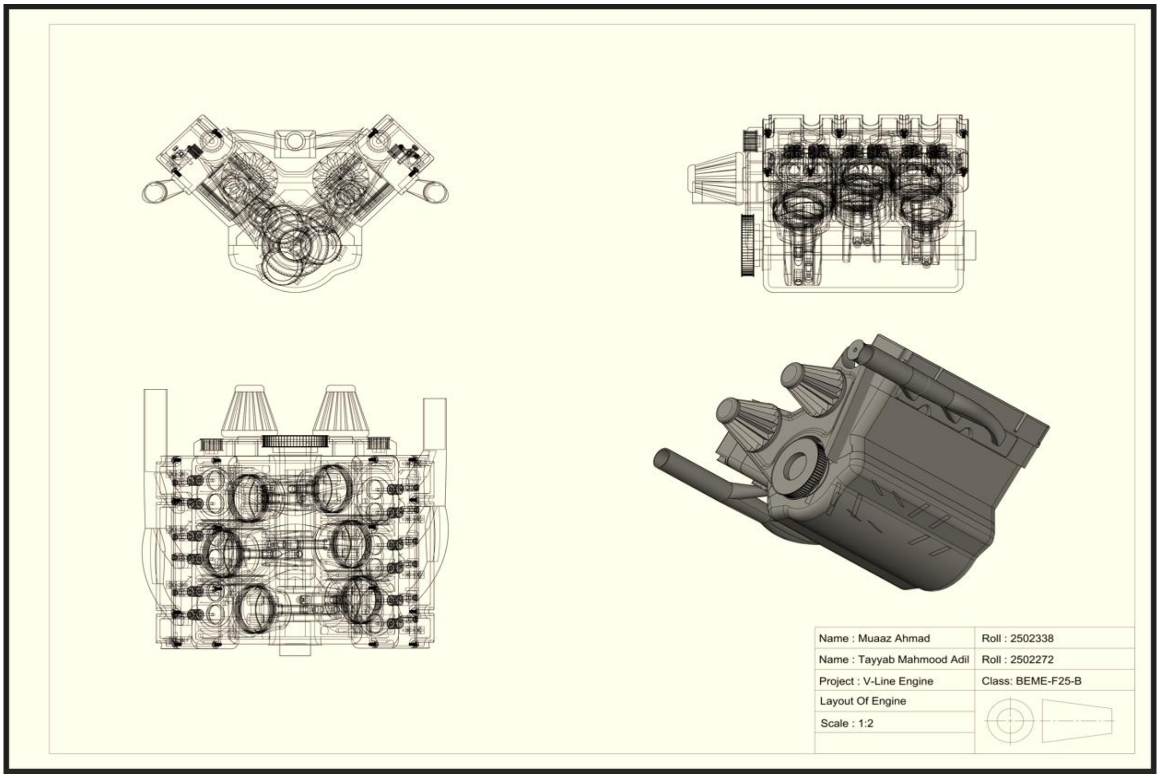
SUBMITTED TO
SIR MEHTAB

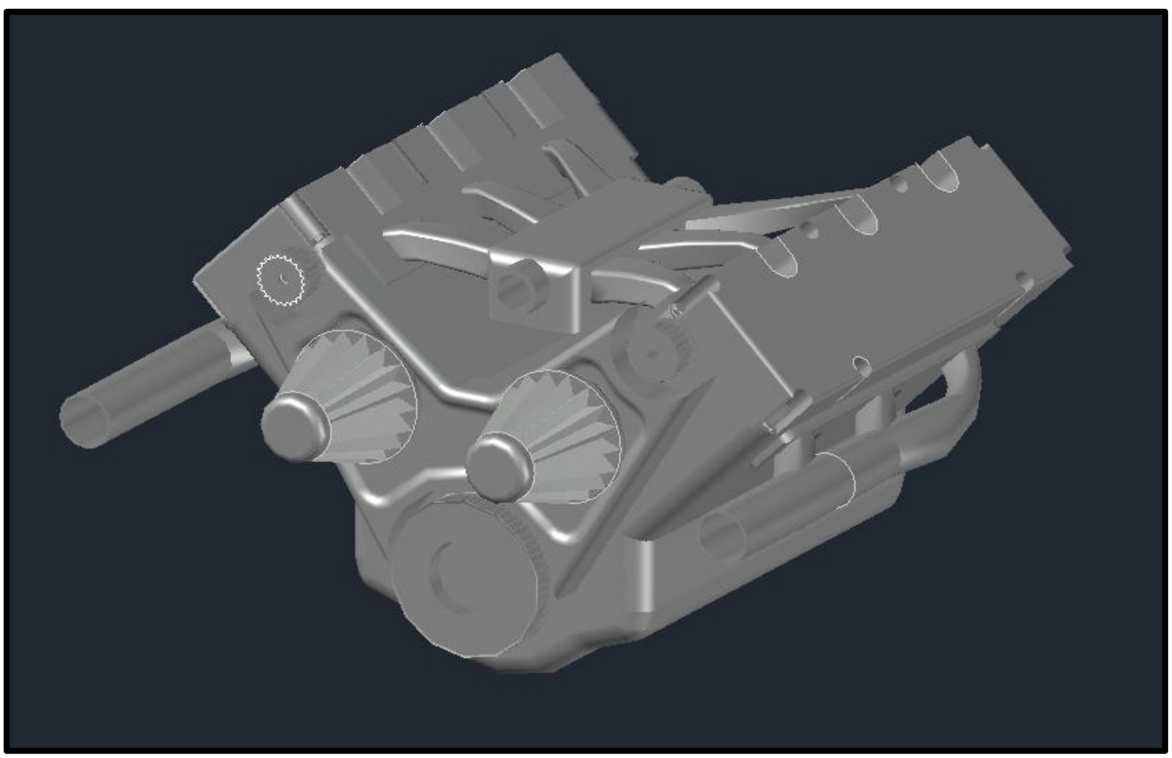





Name : Muaaz Ahmad
Roll : 2502338
Name : Tayyab Mahmood Adil
Roll : 2502272
Project : V-Line Engine
Layout Of Engine
Class: BEME-F25-B
Scale : 1:2

PROJECT OVERVIEW





ABSTRACT

This project presents the design and development of a six-cylinder V-line internal combustion engine. The main purpose of this work is to explore a compact and efficient engine configuration that combines the advantages of inline and V-type engines. Modern vehicles require engines that occupy less space, provide smooth operation, and maintain good performance. The V-line engine layout is studied in this project as a possible solution to these requirements.

The six-cylinder V-line engine is designed with a unique cylinder arrangement where the cylinders are placed at a small angle to each other, forming a V-like structure while maintaining an inline-style crankshaft. This arrangement helps reduce the overall length and width of the engine compared to a traditional inline six-cylinder engine. At the same time, it improves balance and reduces vibration when compared to some conventional engine layouts. The engine follows the basic working principle of a standard internal combustion engine and operates on the four-stroke cycle.

During the design process, important parameters such as cylinder positioning, crankshaft design, firing order, and engine compactness were considered. The focus was kept on simplicity, mechanical feasibility, and ease of understanding rather than high-end performance calculations. The model was developed to clearly show how the components interact and how power is produced inside the engine. The design also highlights how innovative layouts can improve space utilization without making the engine overly complex.

This project helps in understanding engine construction, working principles, and modern design approaches used in automotive and mechanical engineering. The six-cylinder V-line engine demonstrates how alternative configurations can offer practical advantages in terms of size, balance, and efficiency. The knowledge gained from this project is useful for engineering students and can be applied in the development of small vehicles, motorcycles, and compact power units in the future.

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1. INTRODUCTION

Internal combustion engines are an important source of power for vehicles, machines, and many industrial applications. Even today, there is a strong demand for engines that are compact, efficient, and reliable. One of the key factors that affects engine performance and size is the arrangement of cylinders. Because of this, engineers continue to develop new engine configurations that improve space utilization and reduce mechanical issues such as vibration.

This project focuses on the design and study of a six-cylinder V-line engine. The V-line engine is a combination of inline and V-type engine layouts. In this configuration, the cylinders are arranged at a small angle, forming a V shape, while operating with a common crankshaft similar to an inline engine. This design helps reduce engine length and improves balance when compared to a conventional inline six-cylinder engine.

The engine works on the four-stroke internal combustion cycle, where power is produced through the intake, compression, combustion, and exhaust processes. The project mainly emphasizes understanding the construction, working principle, and design logic of the engine rather than complex performance calculations. Through this work, the advantages of the V-line configuration in terms of compactness, smooth operation, and mechanical simplicity are clearly demonstrated.

This project provides practical knowledge of modern engine design and helps in developing a strong foundation in automotive and mechanical engineering. The study of the six-cylinder V-line engine also highlights how innovative engine layouts can meet the requirements of compact and efficient power systems.

1.1. MAJOR PARTS CONTAIN

- Cylinder block with V-line cylinder arrangement
- Pistons
- Connecting rods
- Crankshaft
- Camshaft and valve mechanism
- Intake and exhaust system

1.2. REASON FOR SELECTION

- **Showcase Complex Modeling:** Demonstrates proficiency in advanced tools like **Loft** and **Revolve** for irregular, non-standard shapes.

- **Master Angular Geometry:** Proves the ability to work with inclined planes and manipulate the **User Coordinate System (UCS)** effectively.

- **Validate Assembly Skills:** Highlights the capability to align and mate multiple moving components (pistons, rods, crank) without errors.

- **Demonstrate Precision:** Requires exact dimensional control to ensure functional mechanical fitment and realistic proportions.

1.3. PROCEDURE

- **Initiated** the project by establishing the design parameters for a V6 configuration, opting for a standard **60-degree V-angle** to ensure optimal primary balance and compact packaging`
- **Assumed** critical dimensions for the dual-bank cylinder arrangement, including bore spacing and deck height, in the absence of manufacturer blueprints
- **Calculated** the stroke-to-bore ratio to define the engine's displacement characteristics across all six cylinders
- **Drafted** the 2D profiles of the complex engine block, accounting for the two offset cylinder banks (Bank 1 and Bank 2)
- **Constructed** the main V6 block geometry using the **Loft** and **Extrude** commands, ensuring the correct angular relation between the two cylinder banks
- **Modeled** six individual pistons and wrist pins, maintaining precise dimensional consistency for mass equilibrium
- **Designed** the connecting rods with robust big-end bearings capable of mating to the shared or split crankpins on the crankshaft
- **Fabricated** a complex crankshaft model with four main bearings and three crank throws, offset by 120 degrees to achieve an even firing interval
- **Implemented** split-journal crankpins (if simulating a 90-degree V6) or shared crankpins (for a 60-degree V6) to ensure mechanically accurate timing
- **Executed** Boolean **Subtract** operations to create the six cylinder bores and coolant passages within the solid block
- **Overcame** software limitations regarding the shelling of the complex V-block by generating manual internal cores to define wall thickness
- **Refined** the cylinder head mating surfaces and oil pan rails using **Fillet** and **Chamfer** tools for a machined appearance
- **Imported** all components into a master assembly file, organizing them by layer (e.g., "Moving Parts", "Static Block")
- **Manipulated** the User Coordinate System (UCS) to align the pistons within the angled bores of both the Left and Right banks
- **Positioned** the crankshaft centrally within the crankcase, verifying clearance for the counterweights during rotation
- **Mated** the six connecting rod assemblies to their respective crank journals, ensuring proper offset between opposing cylinders

1. COMMANDS USED

- **CIRCLE:** circle is used for various circular parts used in drawing.
- **LINE:** lines are used for various alignment parts and component.
- **TRIM:** for removing unwanted or extra surface from drawing.
- **FILLET:** fillet is used to bend or curve between two parts.
- **JOIN:** join command is used to join separated parts in a single .
- **MOVE :** moving parts and place them required surface we need.
- **MIRROR:** this command is used to make a duplicate of any part to left or either right side of it **.**
- **REVOLVE:** revolve command is used in revolving of flange yoke.
- **PRESSPULL**: for giving desired height to components not having closed boundary.
- **3D ROTATE:** for viewing the drawing from all sides.
- **SUBTRACT:** it is also use to remove parts by own choice .
- **SHELL:** shell command is used to fix faces according to need or also reduce its surface .
- **Erase:** for deleting various unwanted areas.
- **Pan:** for changing the views.
- **Extrude:** for giving desired height to the components.
- **Copy:** for making multiple copies of the same components.
- **SWEEP:** sweep command is used to make pipe or rod type parts.
- **POLAR:** polar command is used in journal to make twice easily .
- **UNION:** for combining various 3D objects.
- **Disassembly :**For remove union

2. Drawings

Components, Assemblies and Layouts:

Belt Wheel 1 and 2

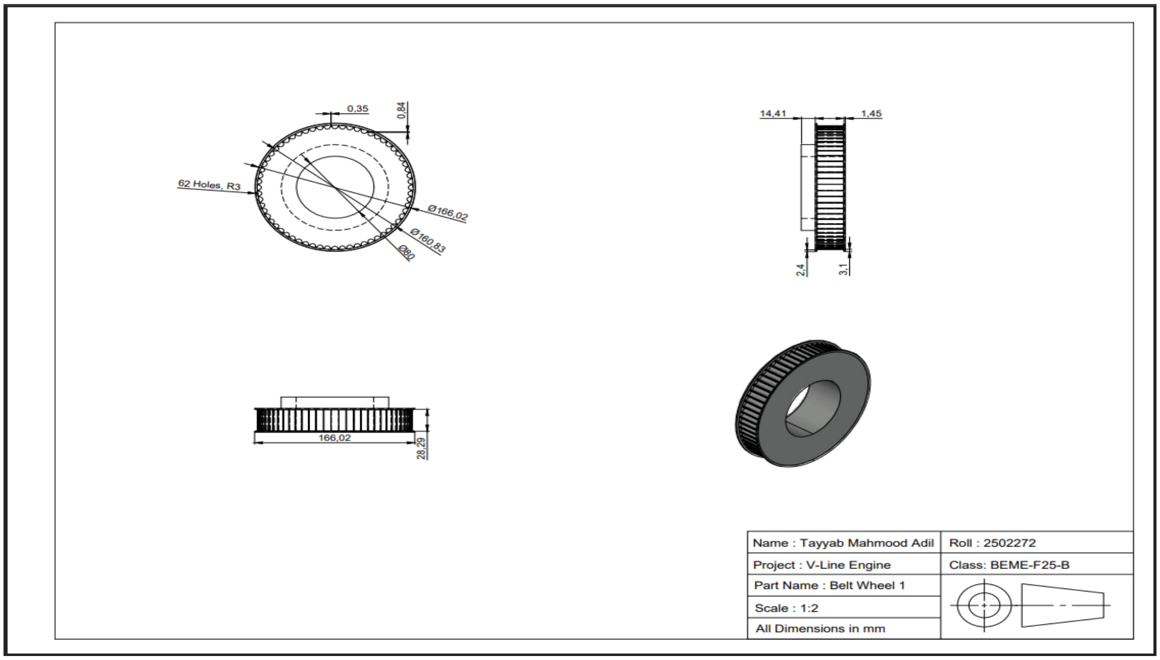


Figure 2(Belt Wheel 1)

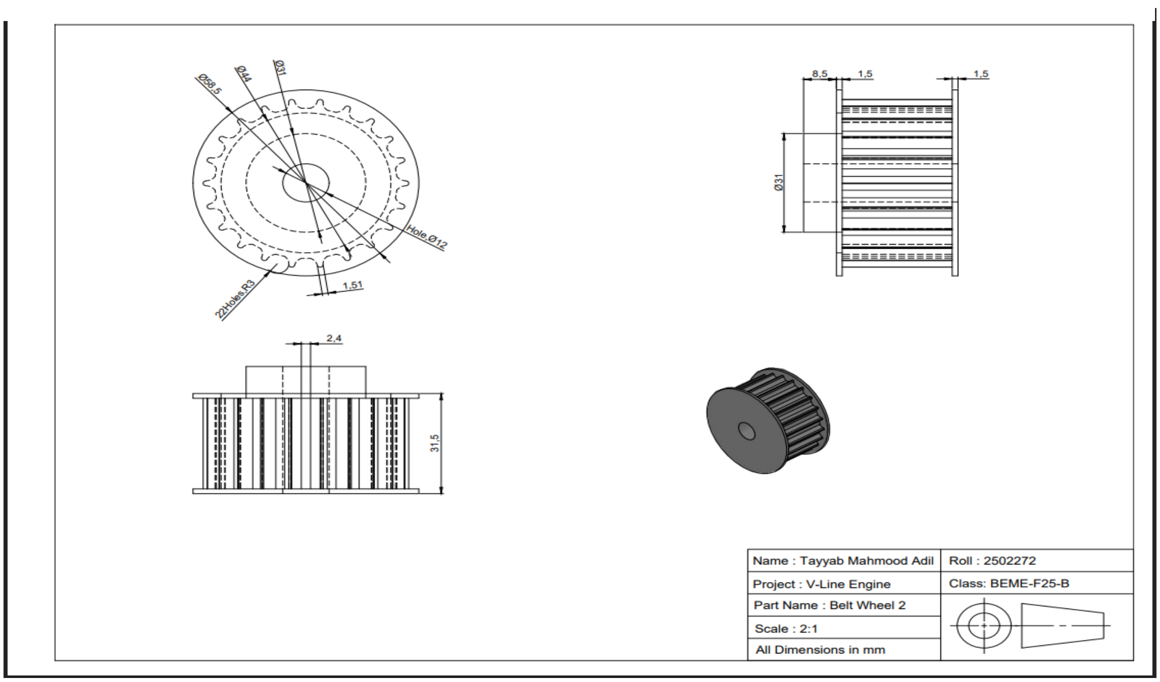


Figure 1(Belt Wheel 2)

Exhaust Manifold

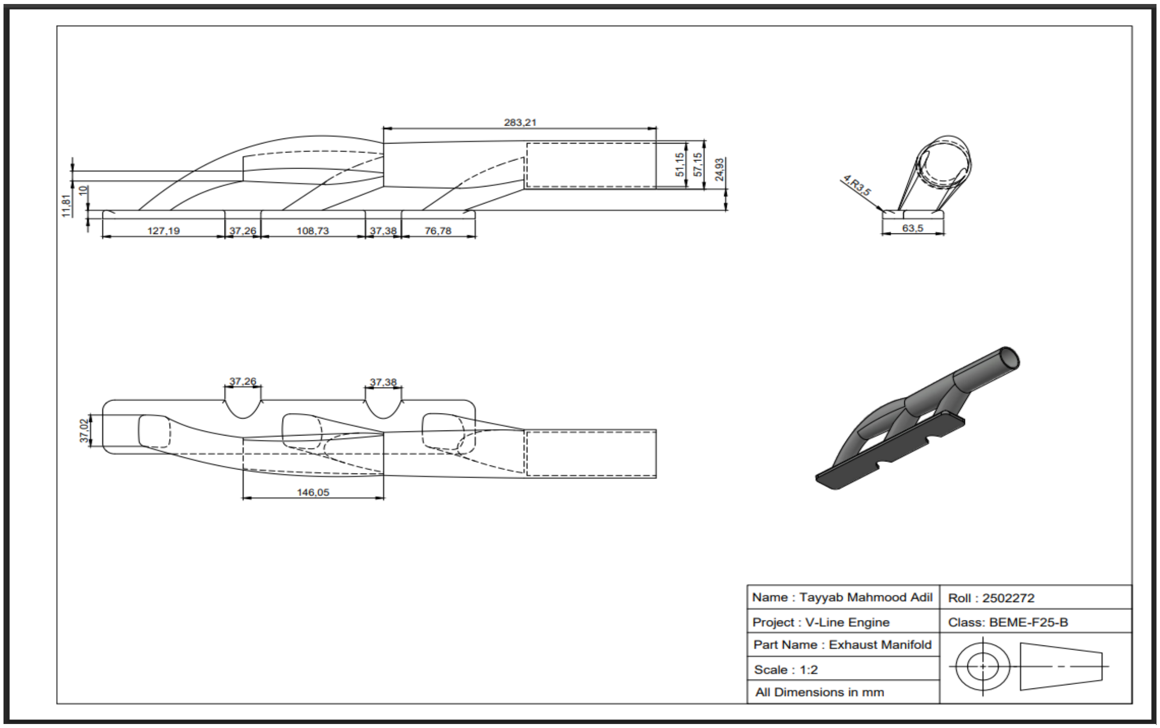


Figure 3(Exhaust Manifold)

Engine Valve Cover

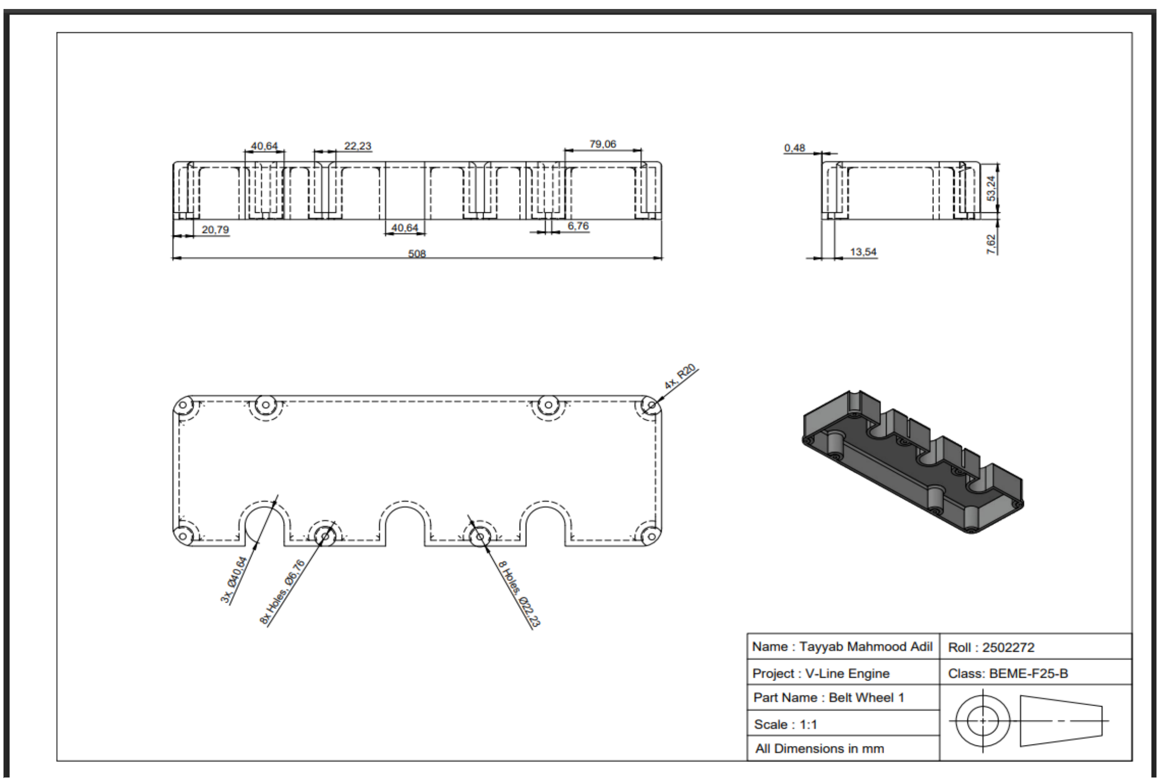


Figure 4 (Valve Cover)

Cylinder Head

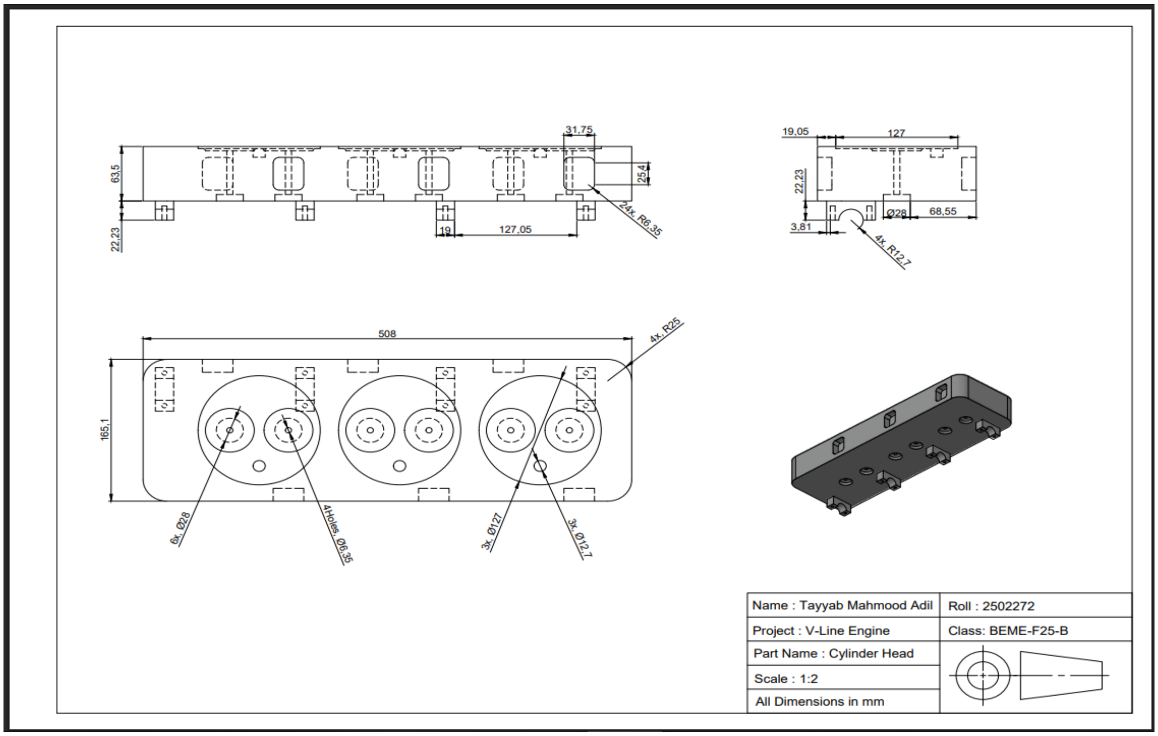


Figure 5 (Cylinder Head)

Camshaft

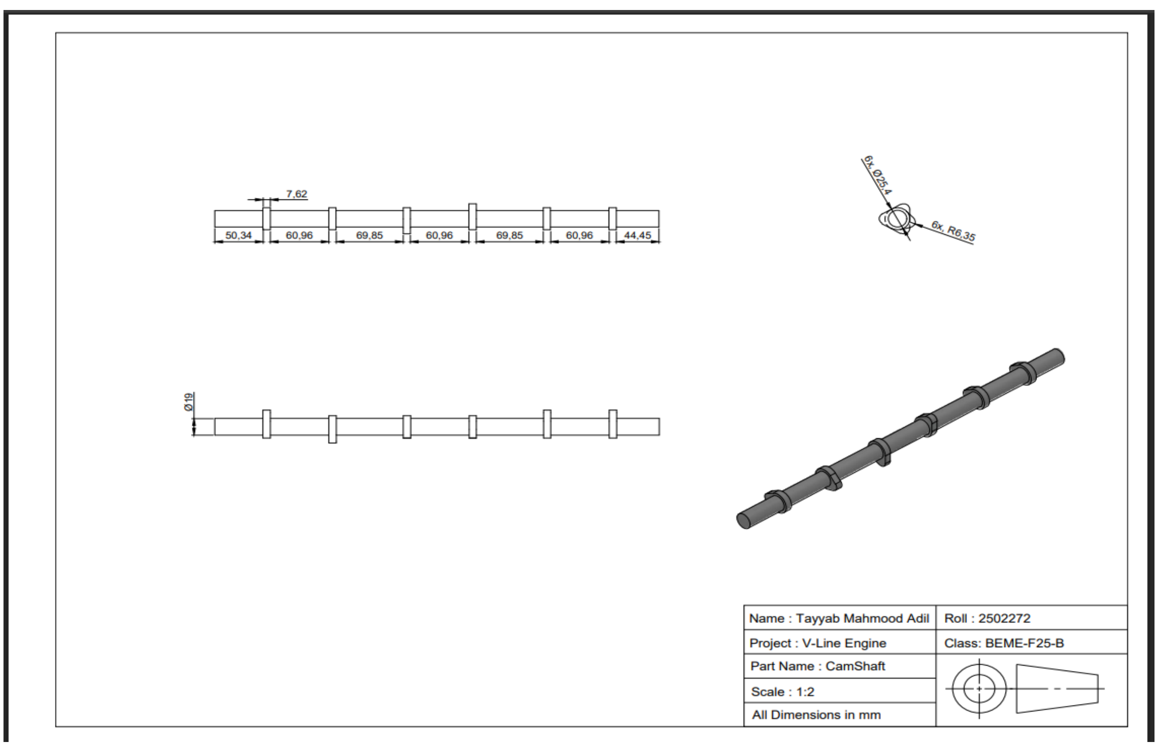


Figure 6 (Camshaft)

Crankshaft

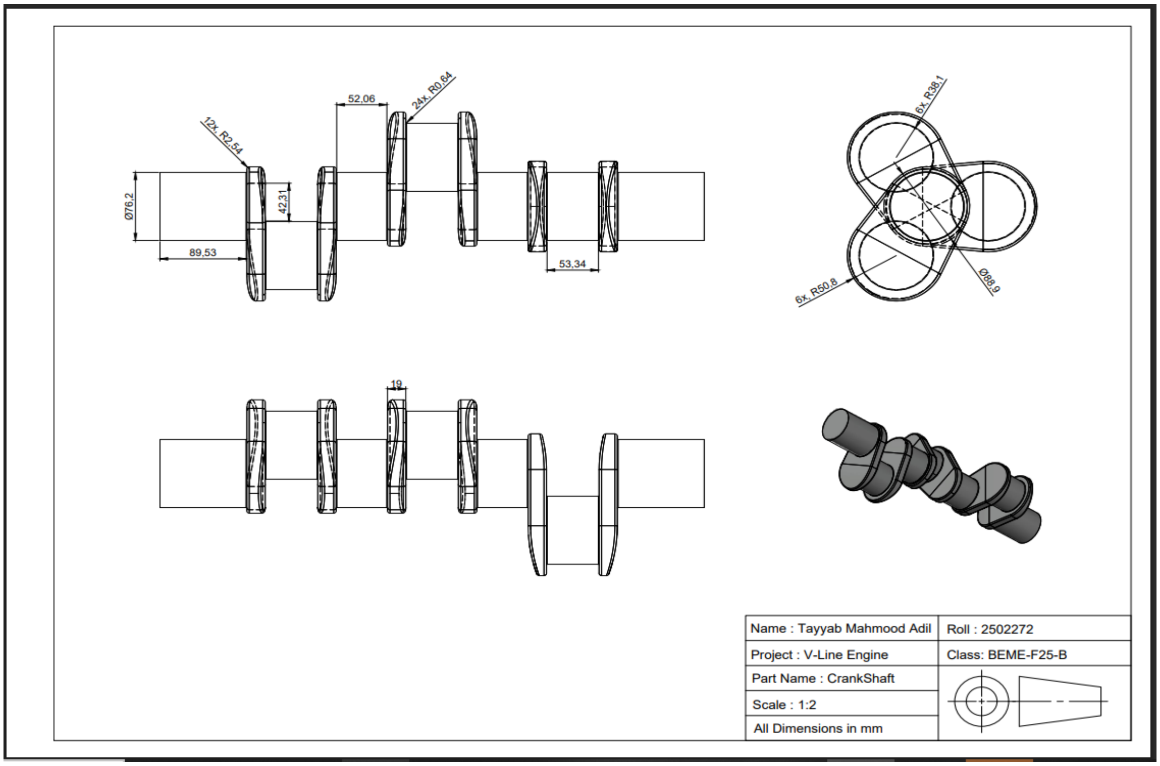


Figure 7 (Crankshaft)

Engine Valve

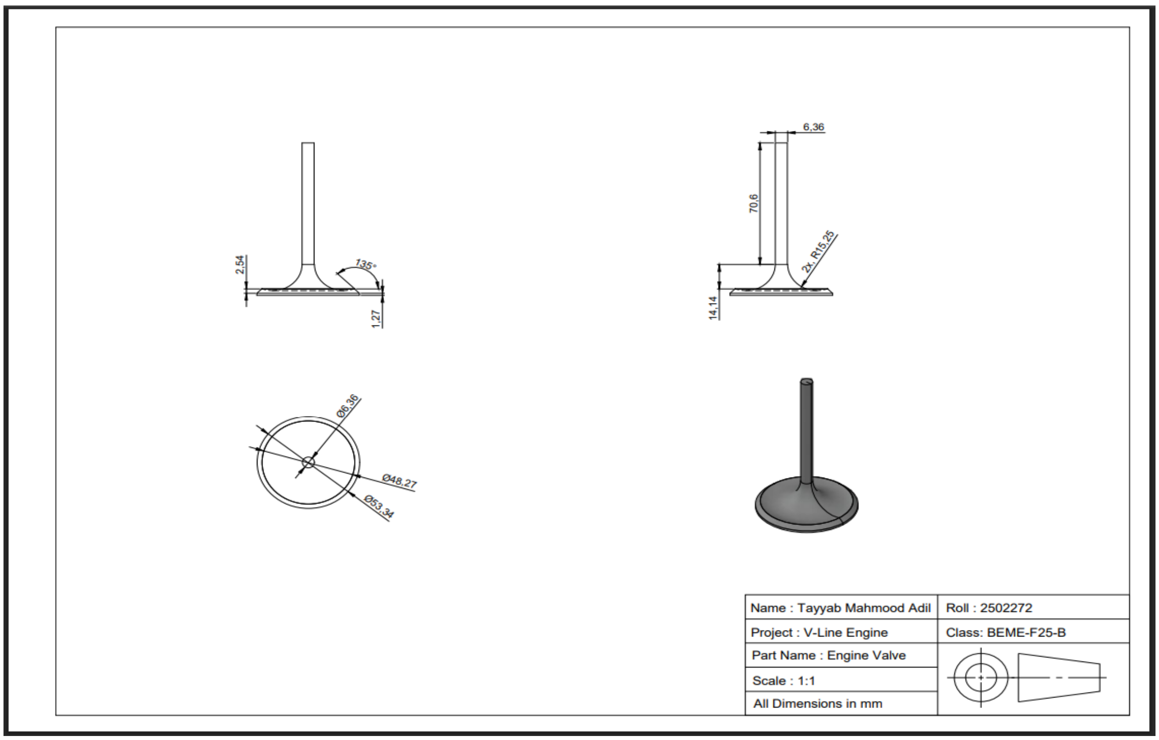


Figure 8 (Engine Valve)

Air Filter

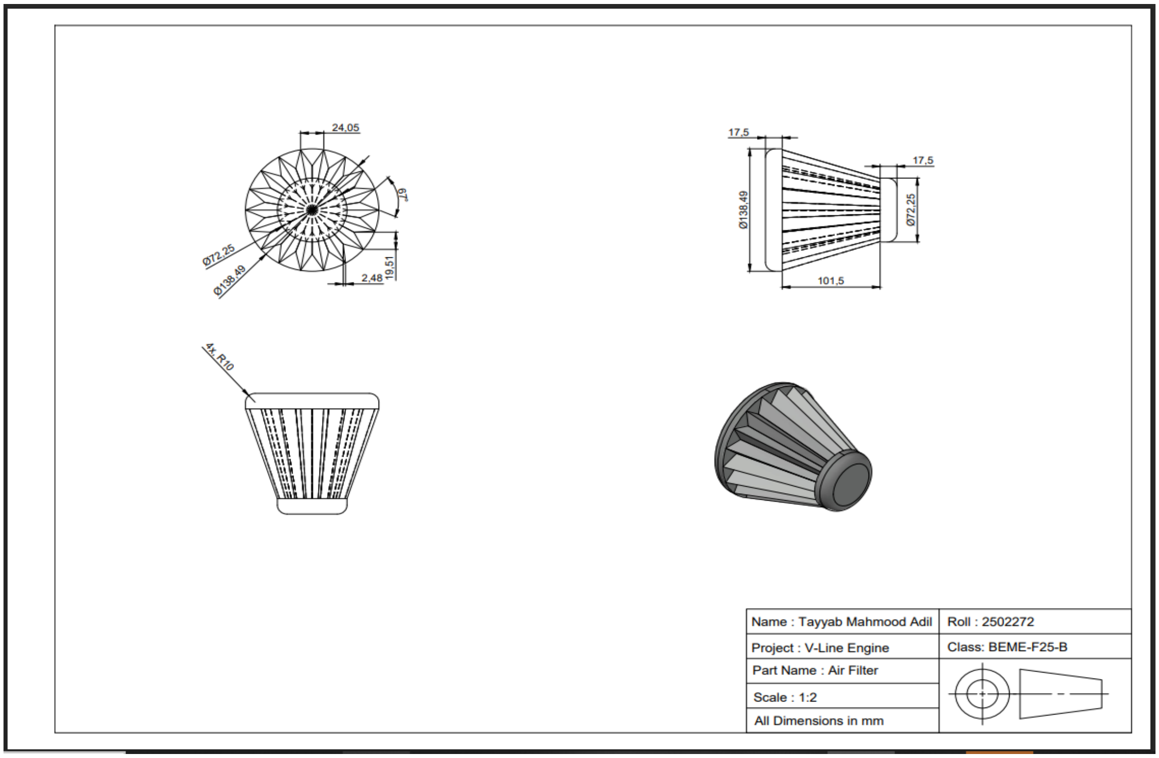


Figure 9 (Air Filter)

Camshaft Retainer

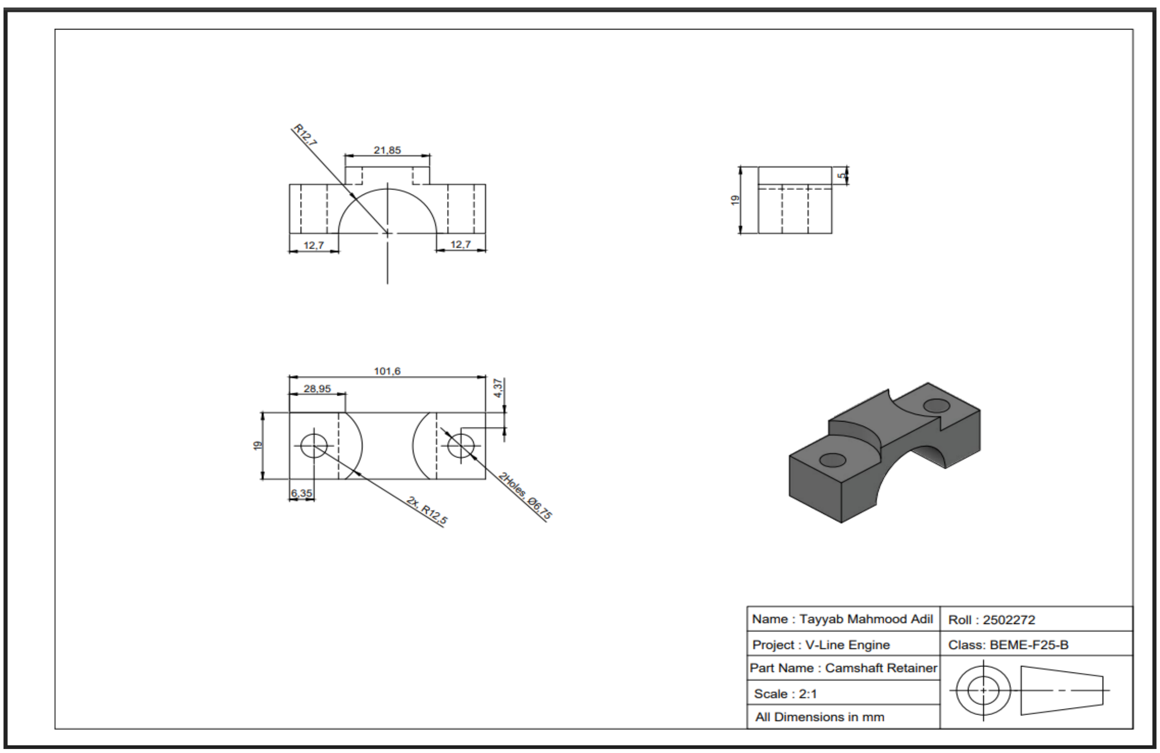


Figure 10 (Camshaft Retainer)

Piston Rod Cap

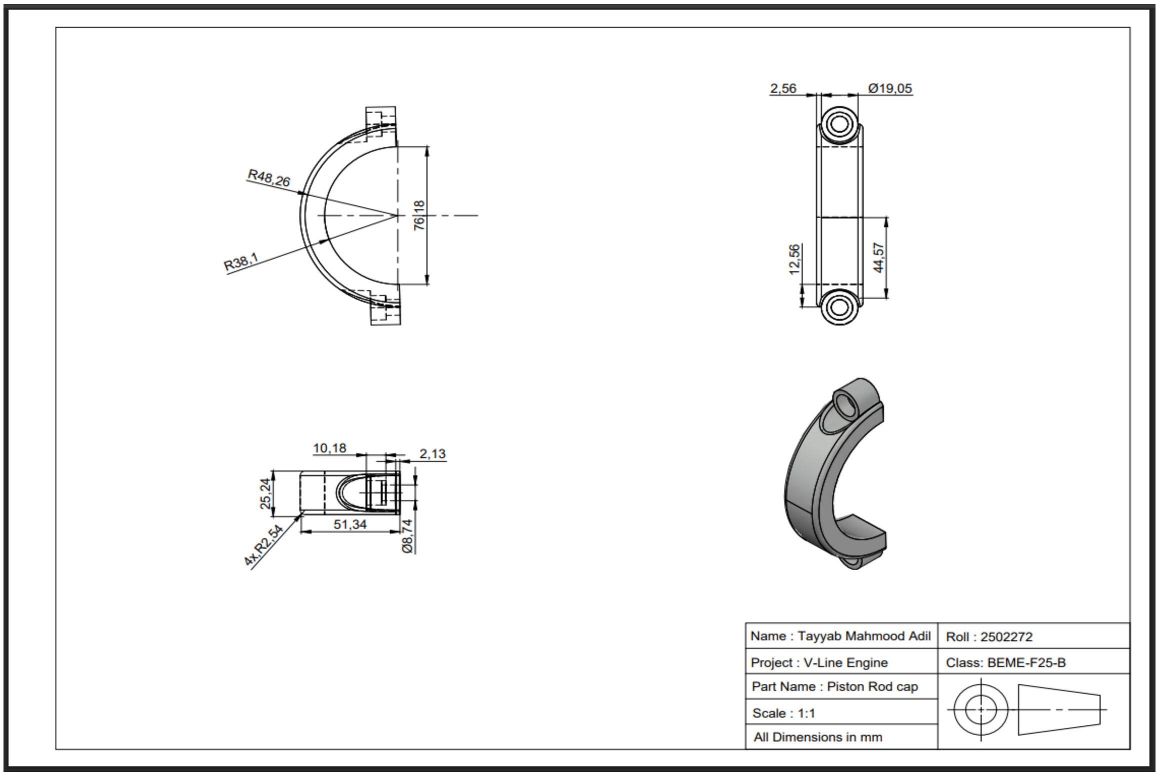
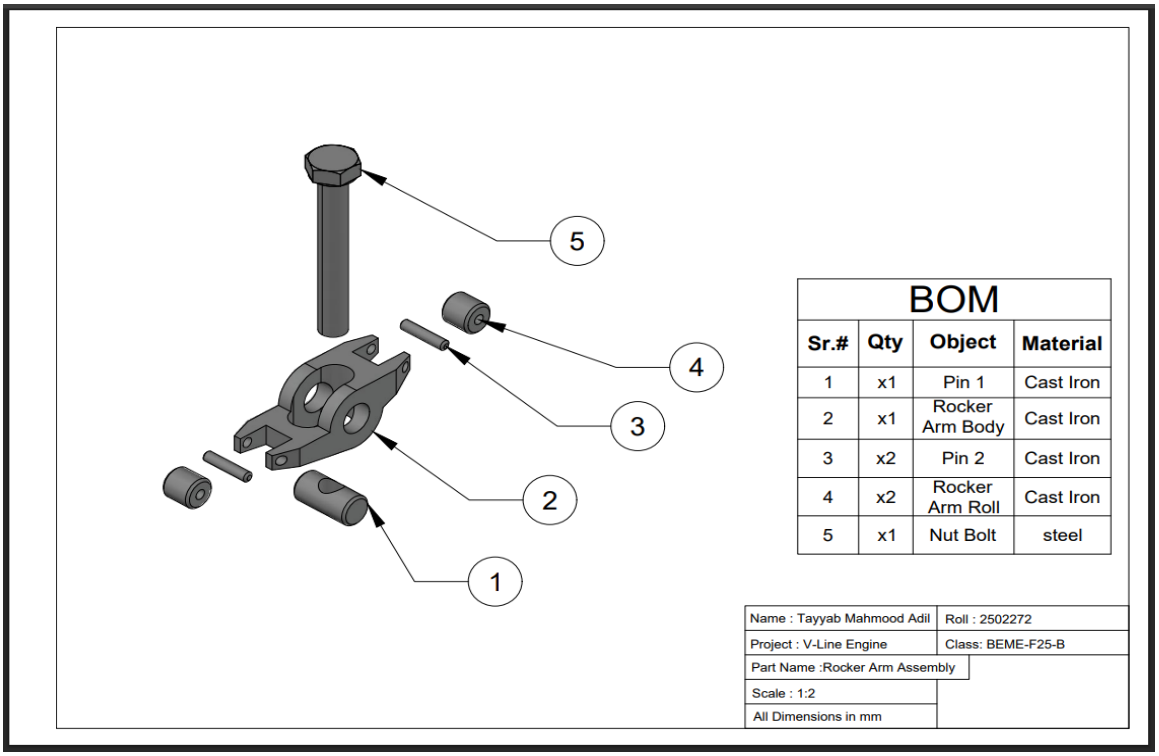


Figure 14 (Connecting Rod Cap)

Rocker Arm Assembly



Sr.#	Qty	Object	Material
1	x1	Pin 1	Cast Iron
2	x1	Rocker Arm Body	Cast Iron
3	x2	Pin 2	Cast Iron
4	x2	Rocker Arm Roll	Cast Iron
5	x1	Nut Bolt	steel

Figure 15 Rocker Arm Assembly

Piston Pin

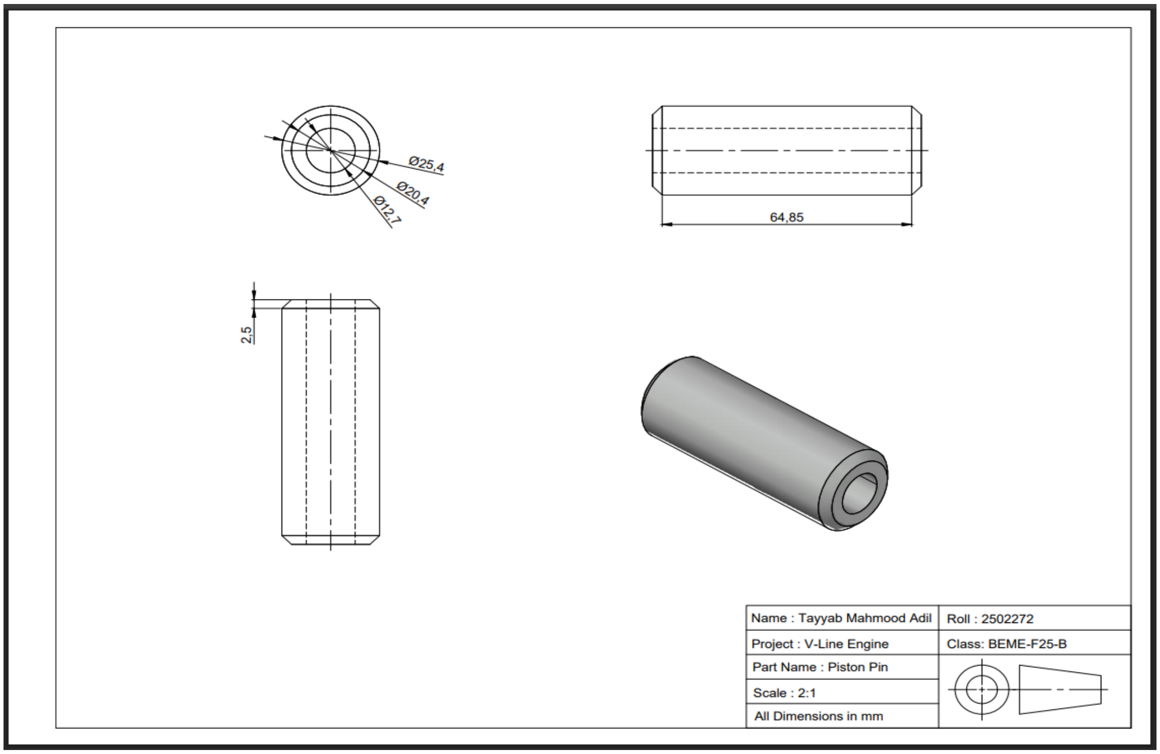


Figure 19(Piston Pin)

M8 Bolt

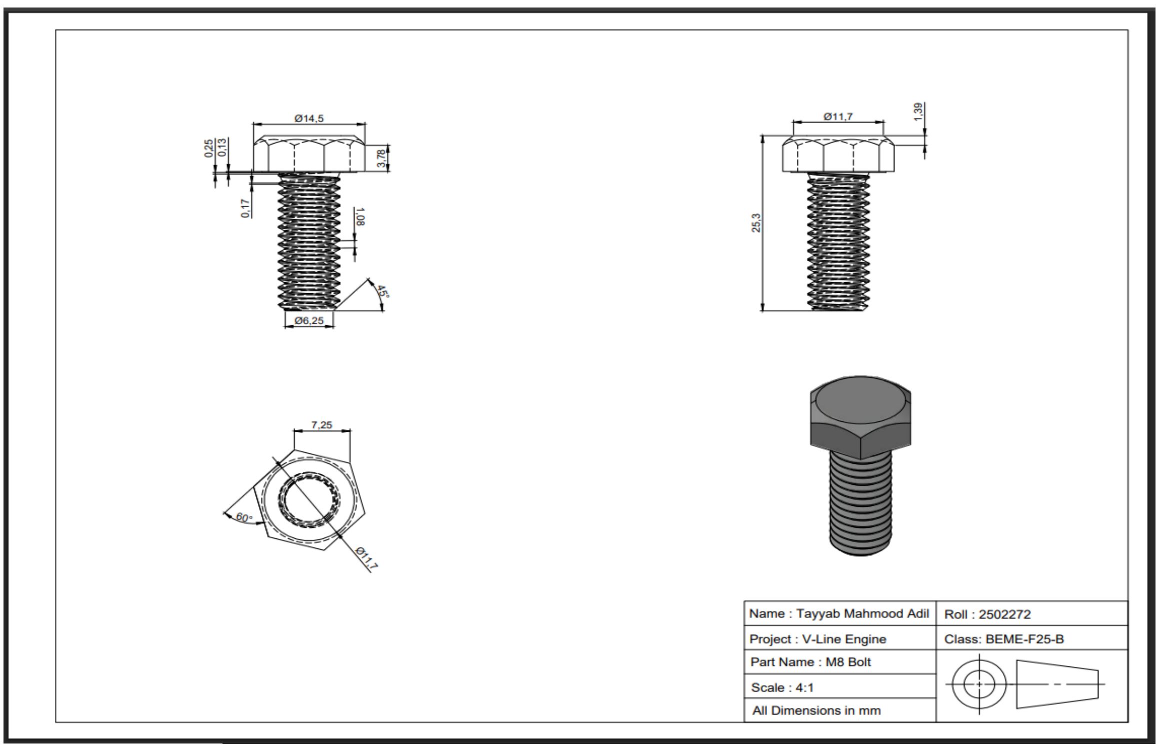


Figure 20 (M8 Bolt)

Oil Pan

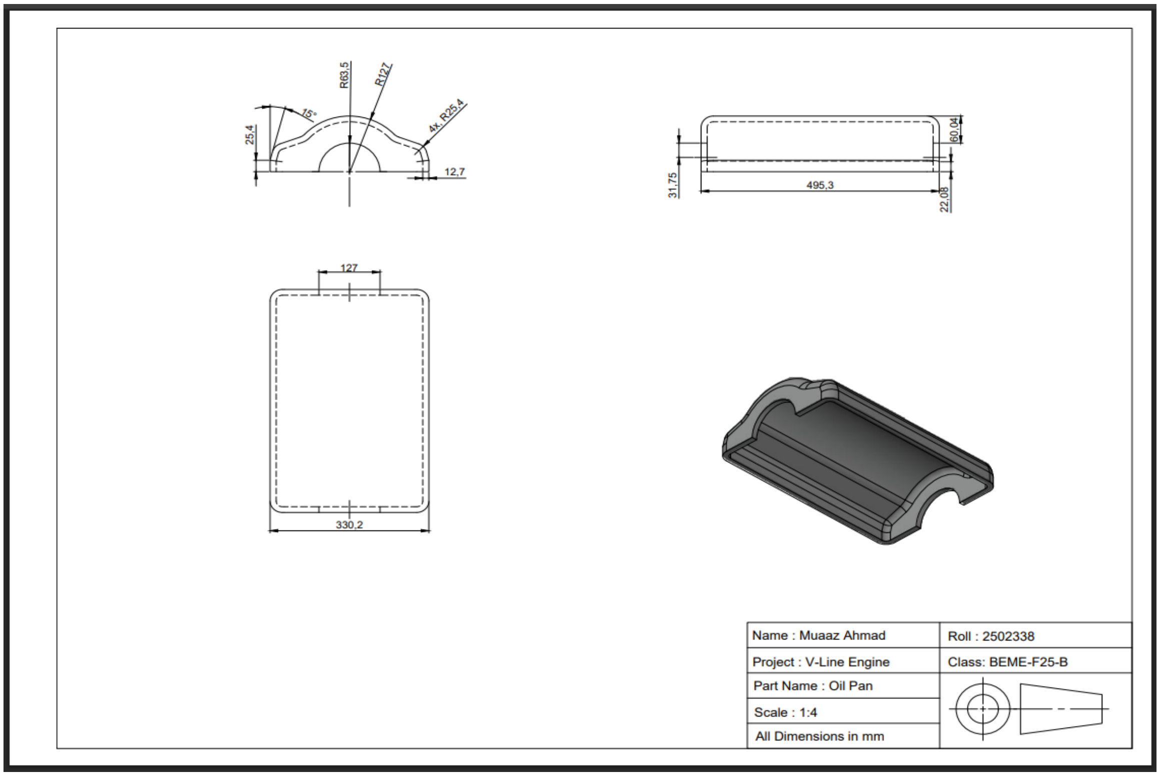


Figure 24 (Oil Pan)

Engine Block

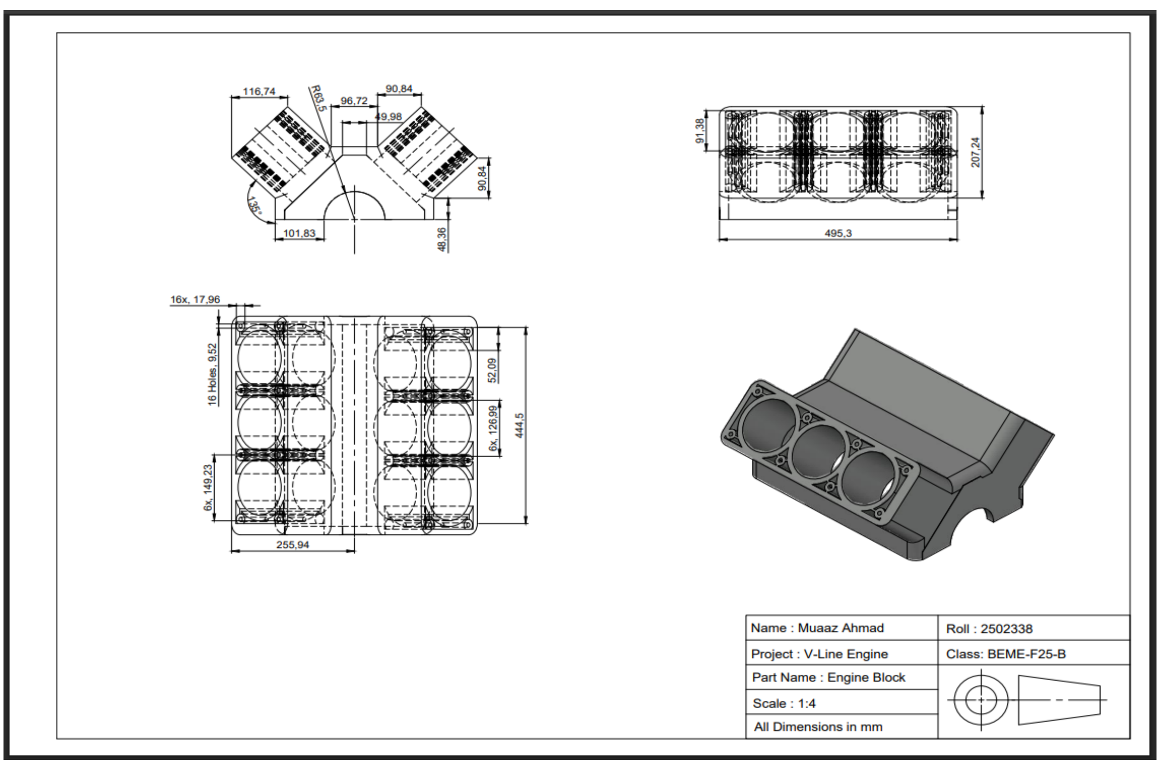


Figure 25 (Engine Block)

Connecting Rod

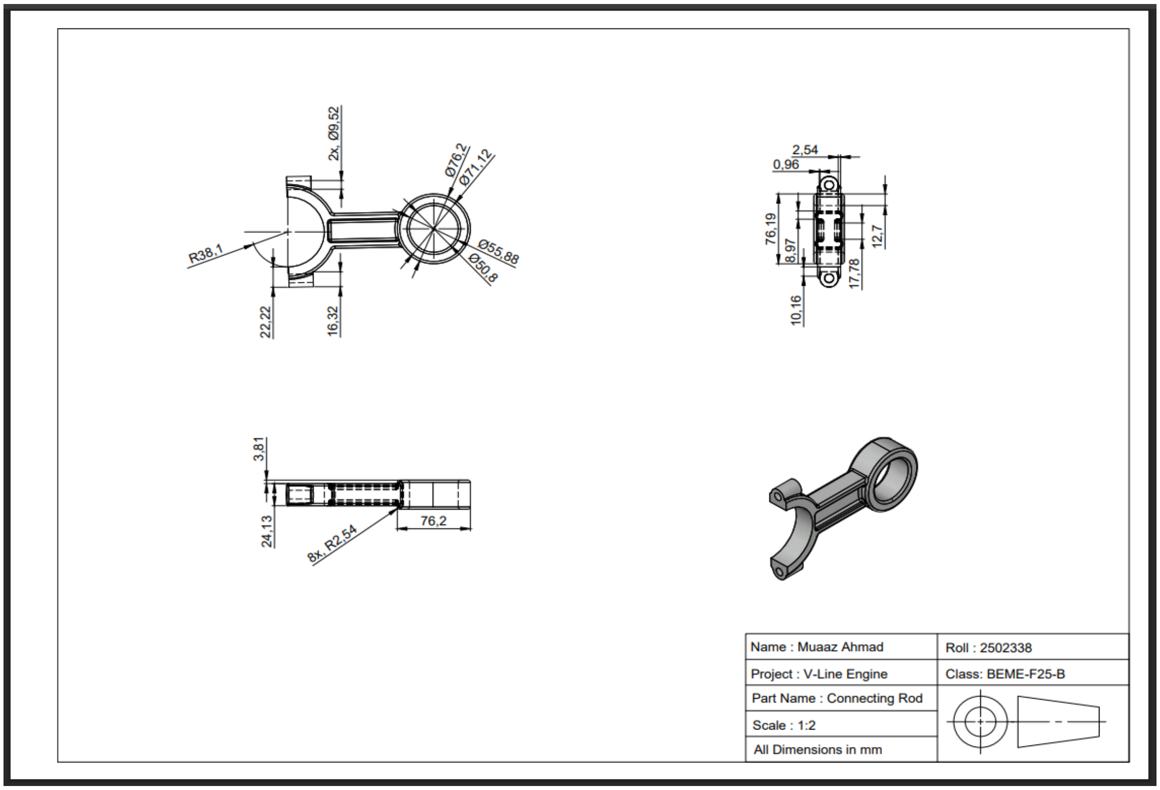


Figure 29 (Connecting Rod)

Piston Head

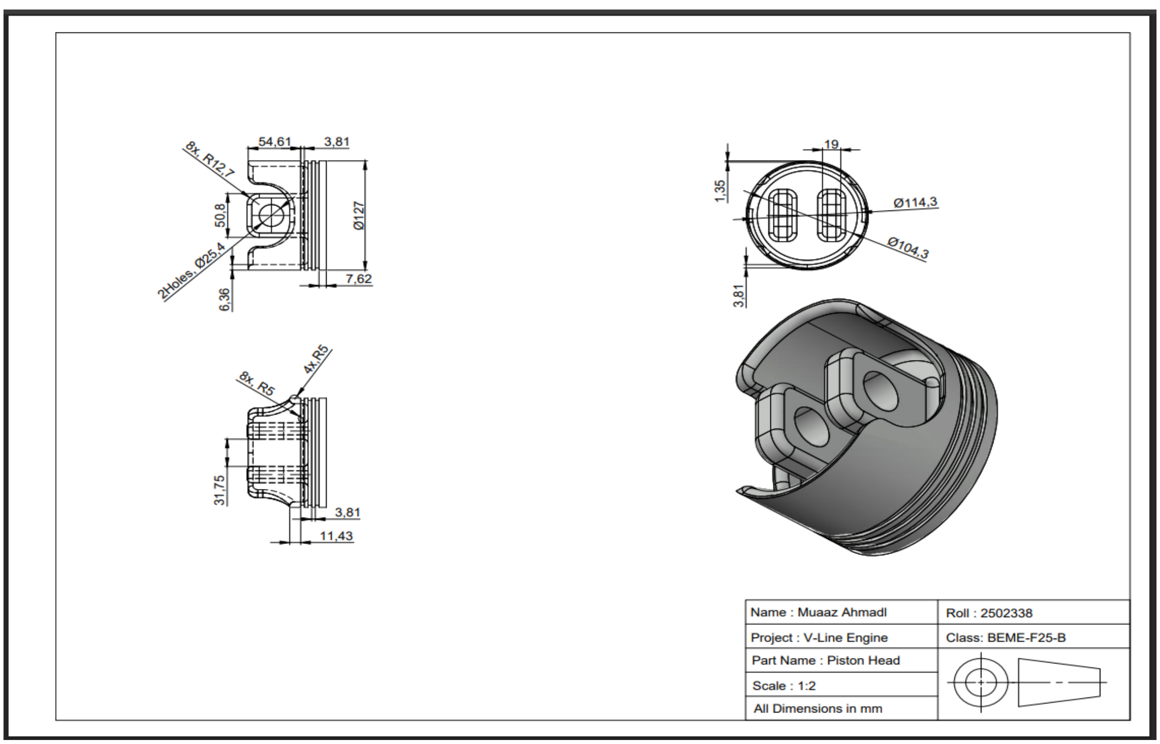


Figure 33 (Piston Head)

Piston Assembly

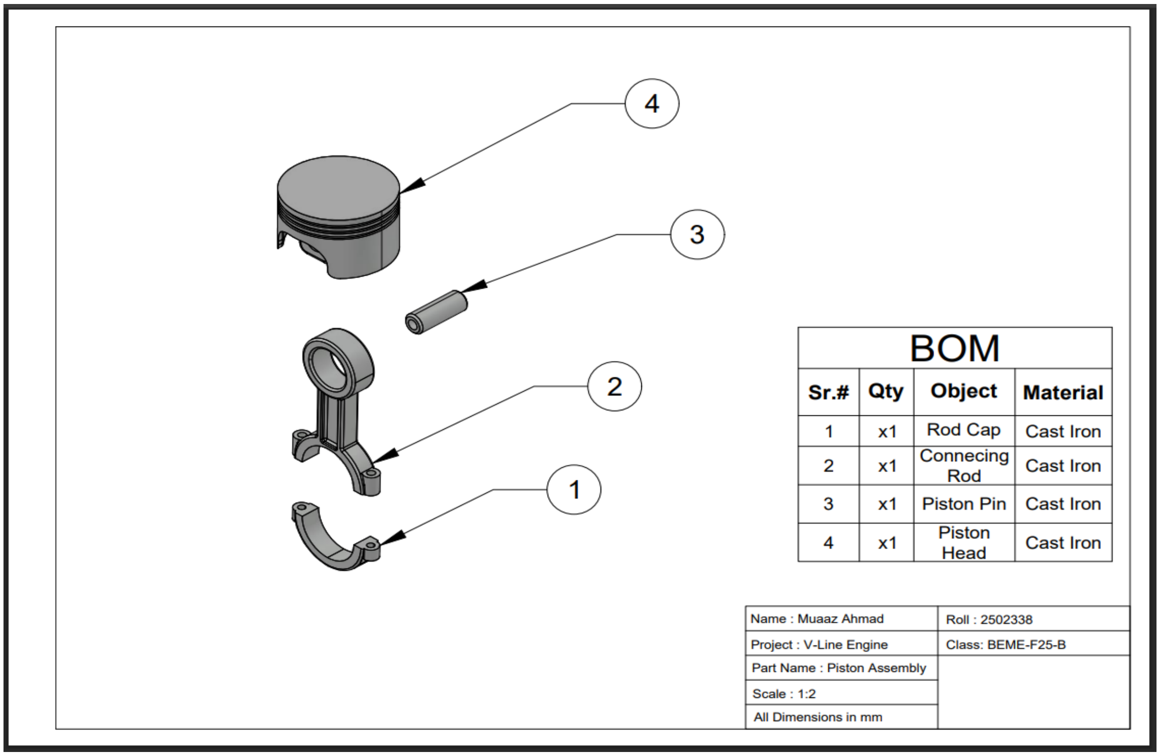


Figure 34 Piston Exploded Assembly

Nut Bolt ISO 8676

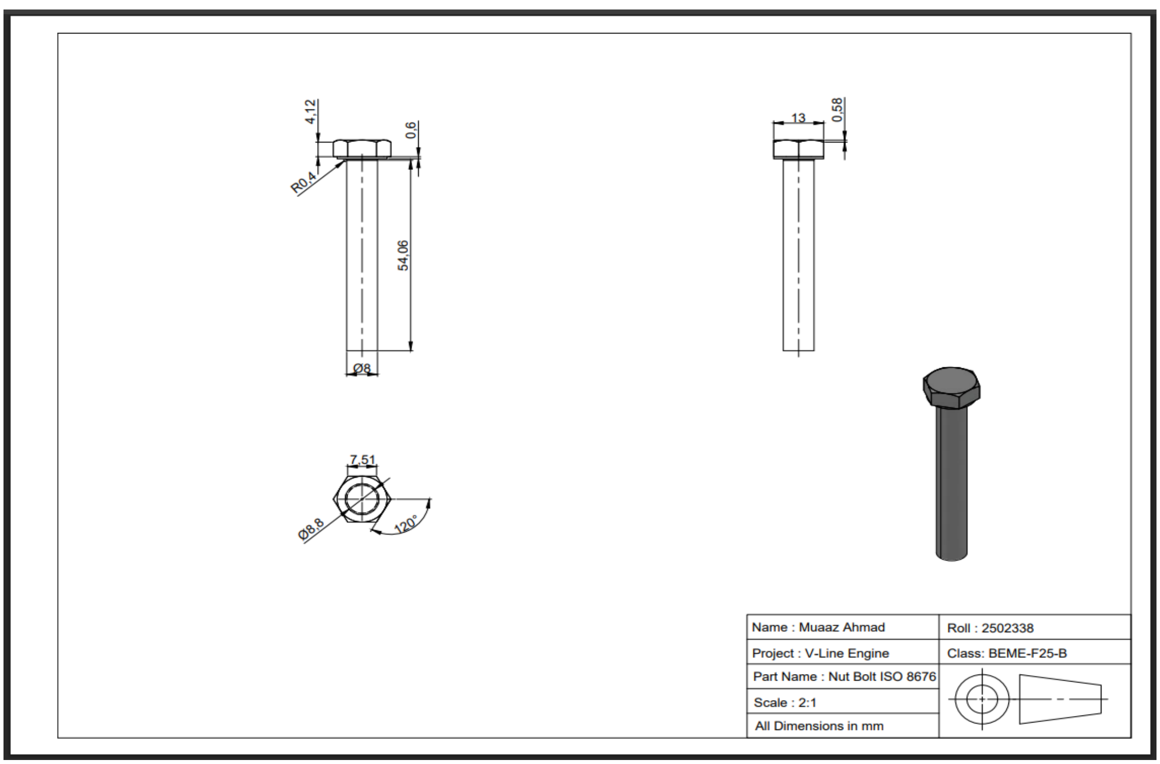


Figure 35 (Iso 8676 Bolt)

Rocker Arm Pin 2

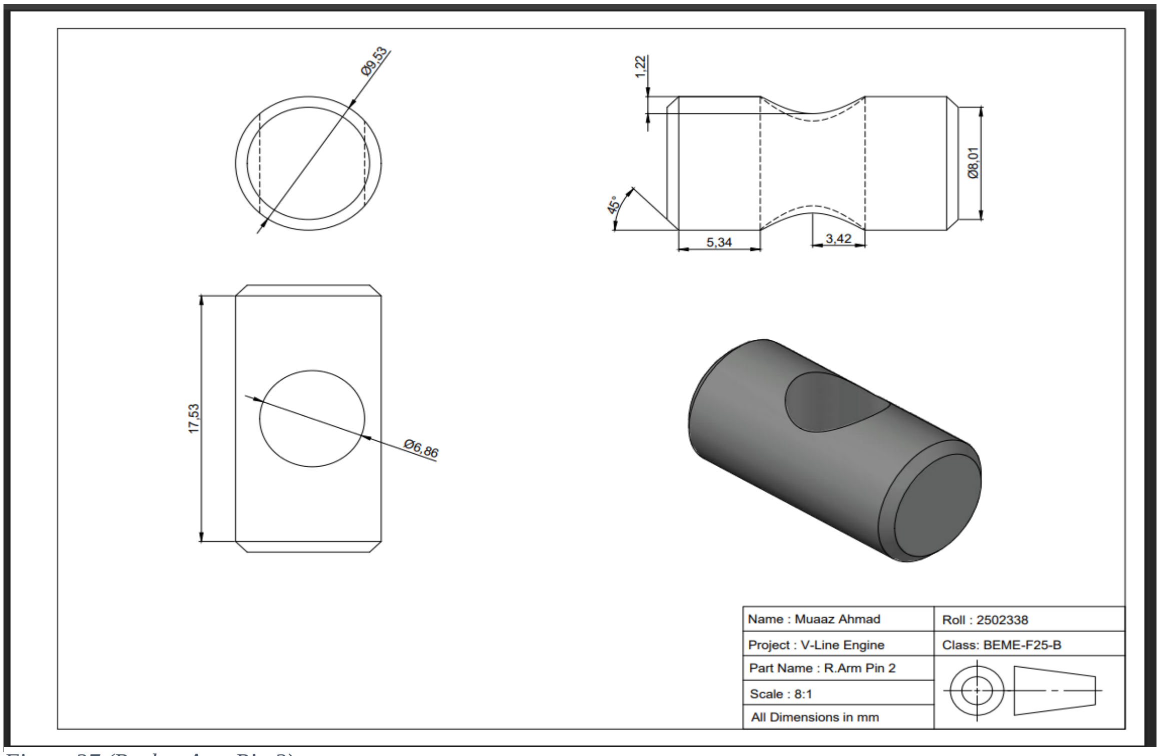


Figure 37 (Rocker Arm Pin 2)

Rocker Arm Pin

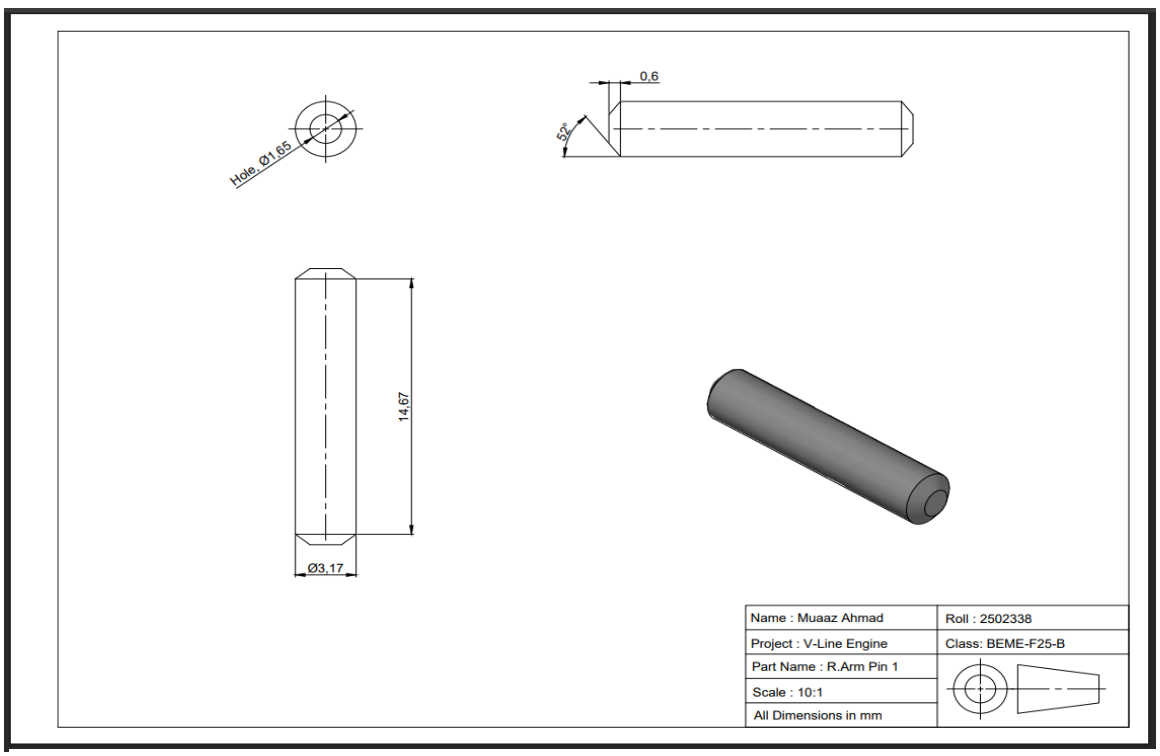


Figure 38 (Rocker Arm Pin 1)

Rocker Arm

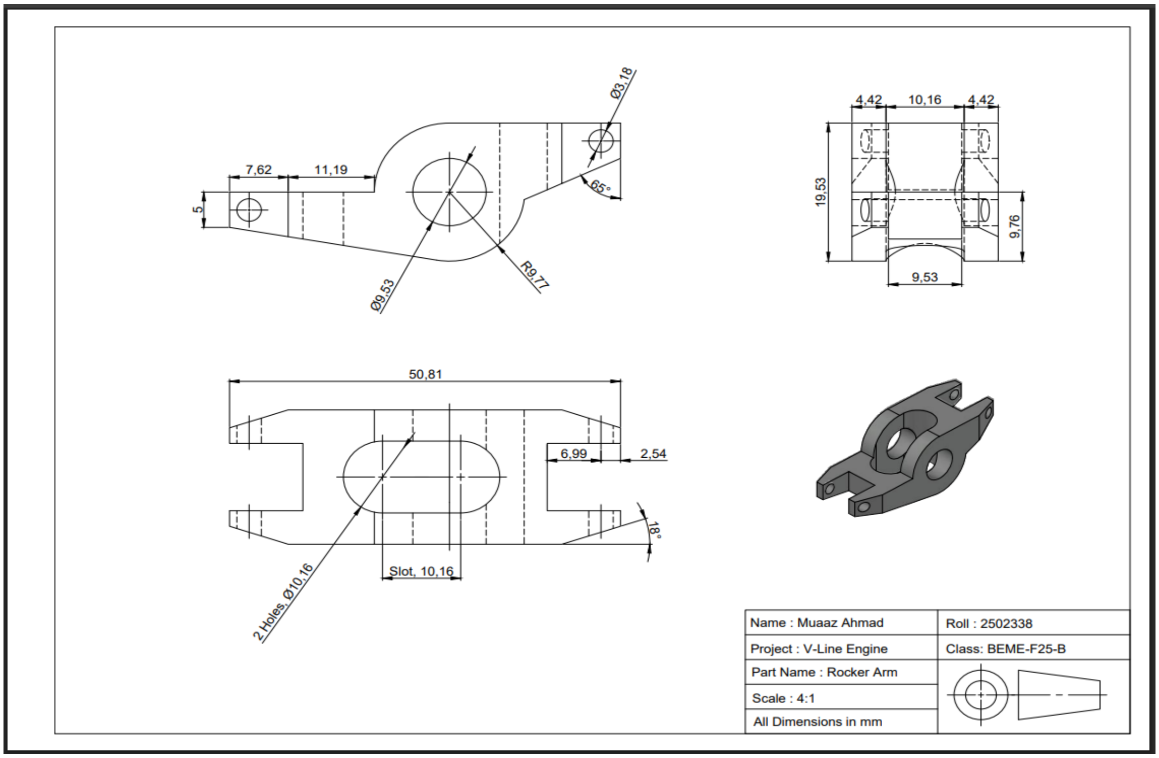


Figure 39 (Rocker Arm Main Body)

Rocker Spring

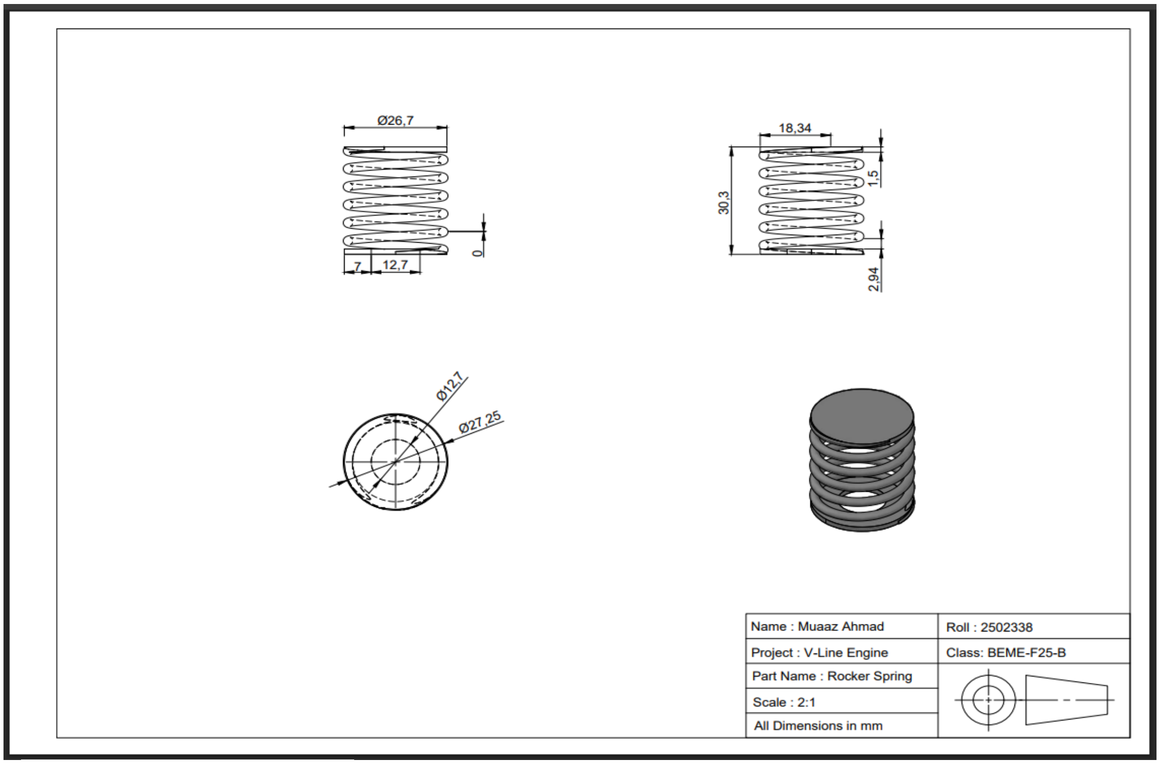


Figure 40 (Rocker Arm spring)

Cam Shaft Bush

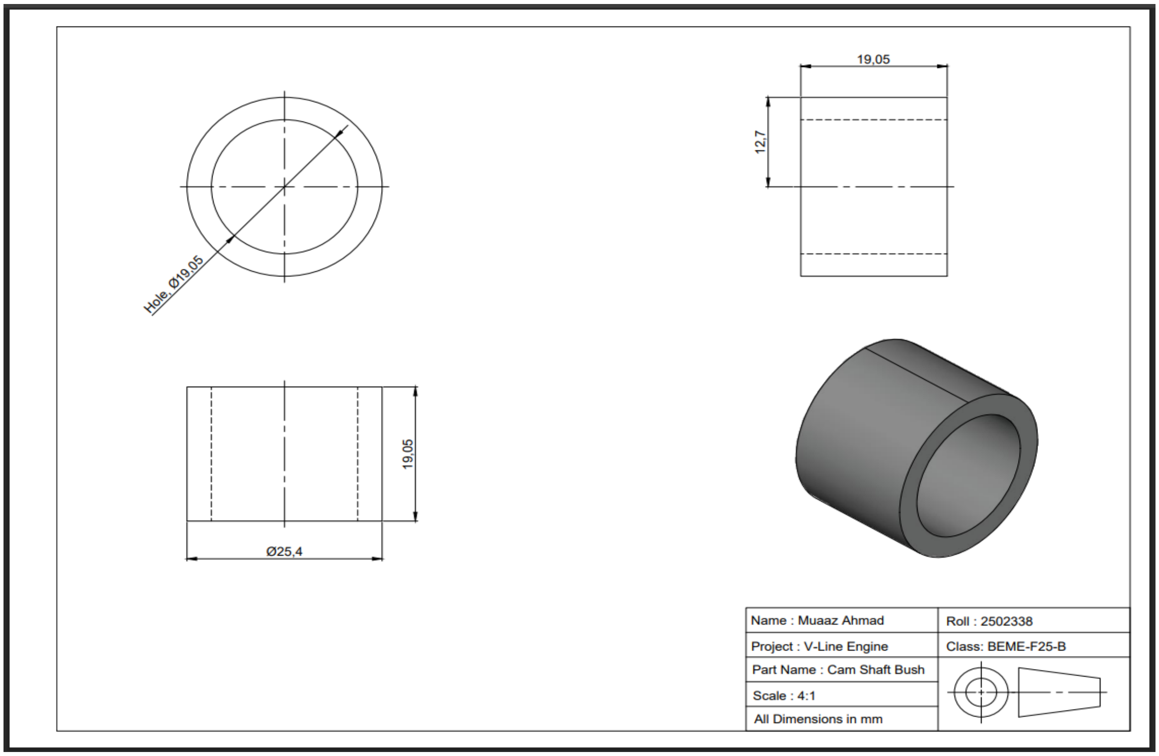


Figure 42 (Camshaft Bush)

Intake Manifold

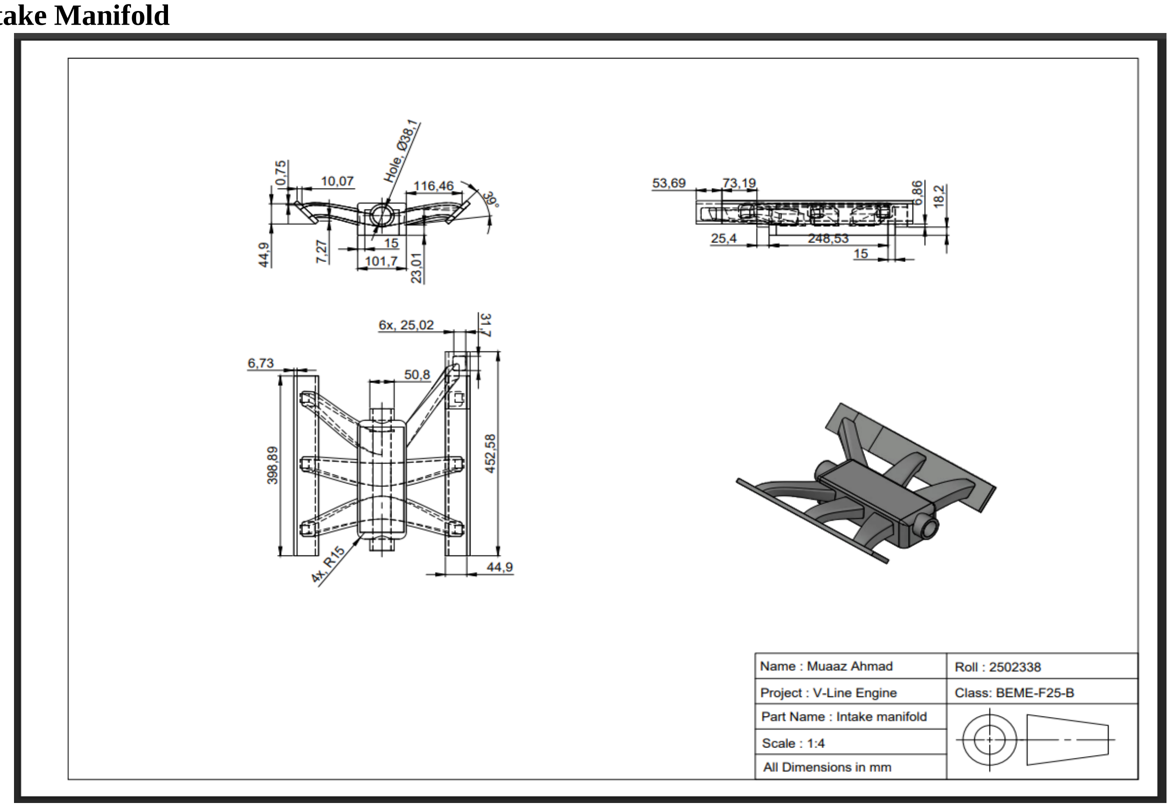


Figure 43 (Engine Intake Manifold)

Crank Shaft Bush

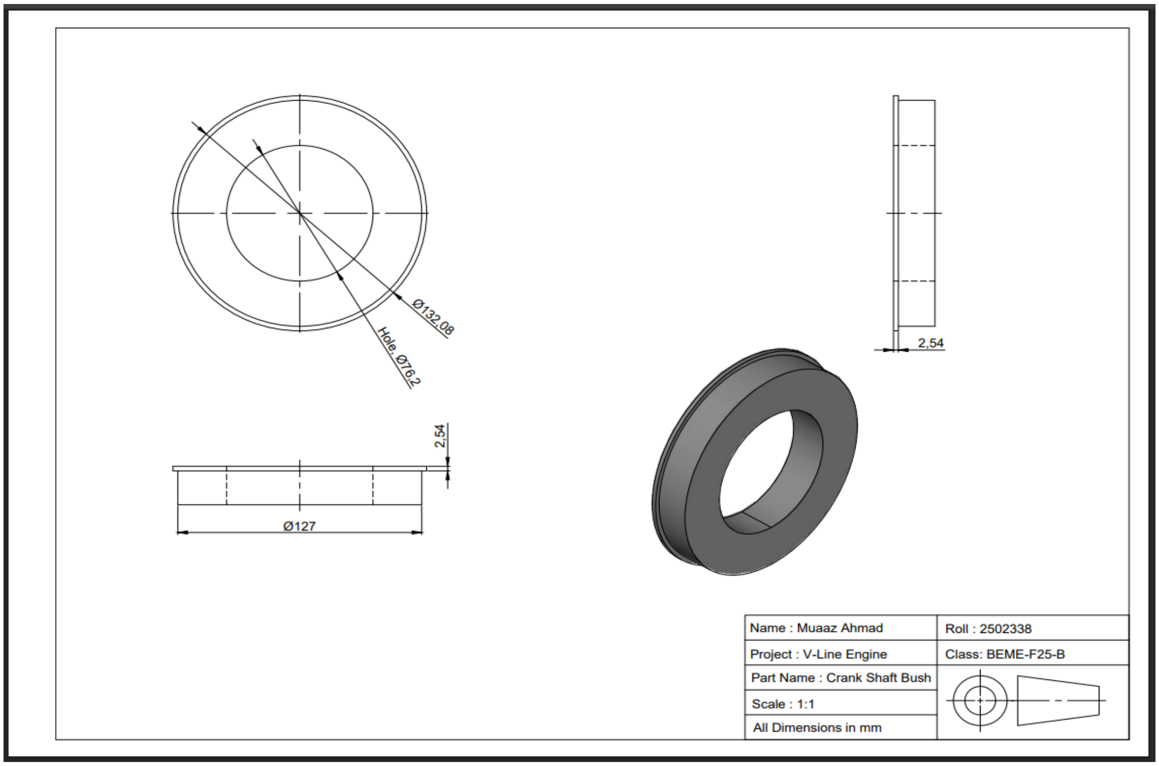


Figure 44(Crankshaft Bush)

Engine Side Box

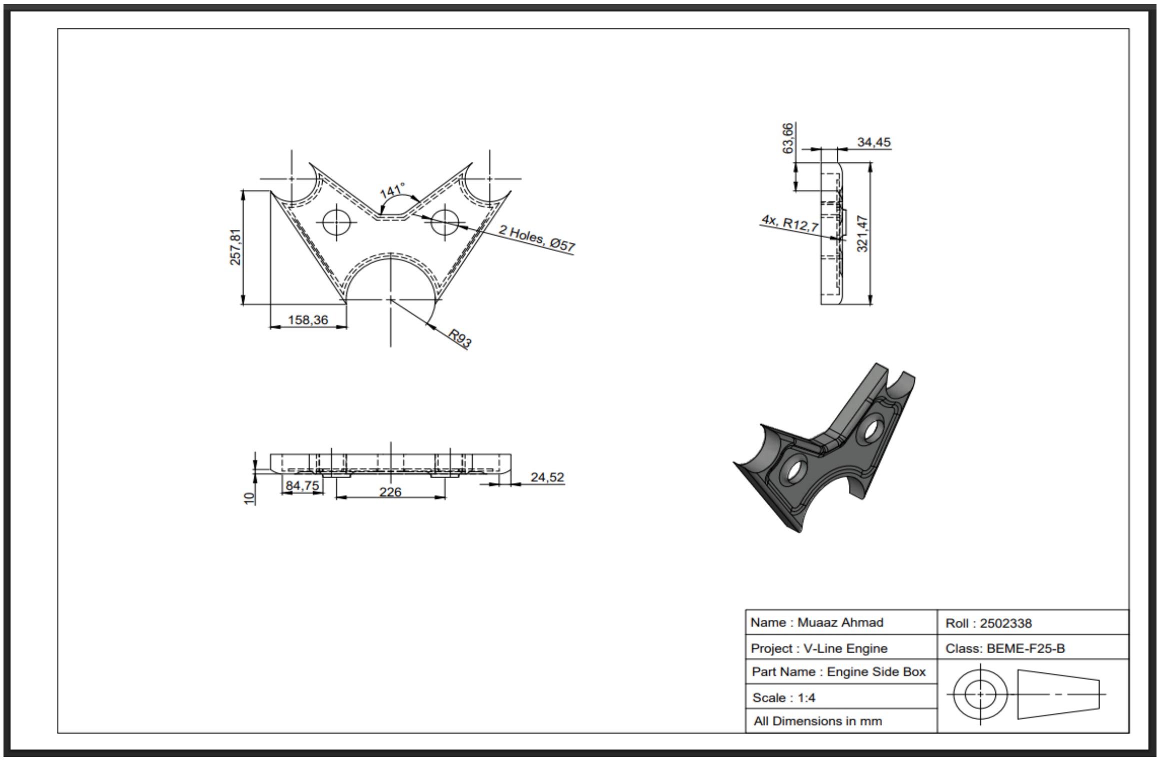


Figure 45 (Engine Side Box)

Final Assembly

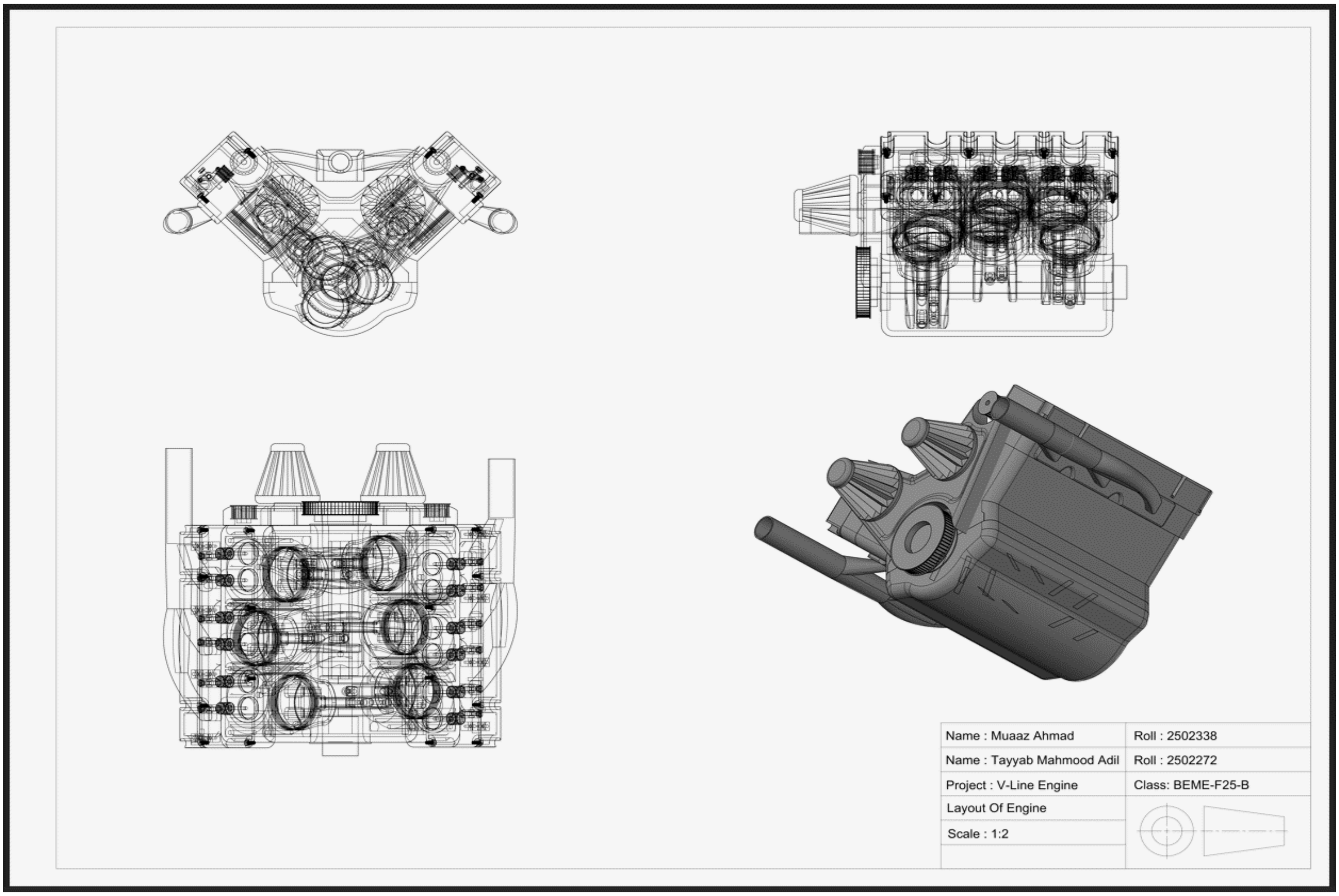


Figure 46 (FINAL ASSEMBLY)

3. PROBLEM FACED

- **Software Limitations**
  A significant technical constraint arose during the modeling phase involving the CAD software's feature set. Specifically, the application of the "Shell" command proved incompatible with complex lofted objects. The software was unable to automatically generate hollow interiors for these irregular shapes, resulting in geometry errors. To resolve this, alternative modeling techniques—such as Boolean subtraction using manually created inner cores—had to be employed to achieve the desired wall thickness.

- **Performance and Rendering Optimization**
  As the complexity of the assembly increased with the addition of detailed components (such as piston, engine block and the crankshaft), the computational load on the hardware intensified. This resulted in lag during the rotation and inspection of the model.

- **Assembly Alignment Issues**
  The most critical challenge occurred during the final assembly phase. Aligning the various moving parts—specifically mating the pistons and connecting rods with the crankshaft—proved difficult. Small calculation errors led to misalignments where components would not snap to their correct center points, requiring extensive troubleshooting.

- **File Management Issues**:
  Losing files or overwriting due to improper saving practices.

- **Incorrect Scaling**:
  Drawing elements at the wrong scale, causing inconsistencies.

- **Layer Mismanagement**:
  Overcrowded or improperly named layers lead to confusion.

4. Work Distribution

Tayyab Mahmood Adil-2502272	• 12 Components Designed • Prepared Report of PBL
Muaaz Ahmad - 2502338	• 14 Components Designed • Prepared 2 Assemblies & Layouts

5. Learnings

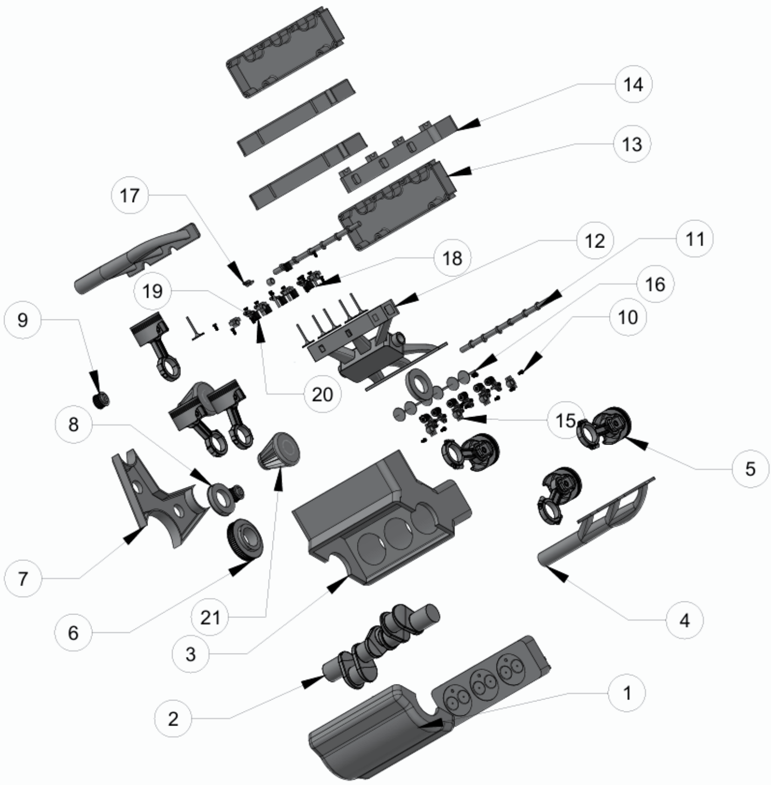
The execution of this V-line engine project served as a pivotal exercise in bridging theoretical engineering concepts with practical digital design. Through the rigorous process of modeling complex mechanical systems, we have derived the following key technical and professional learnings:

- **Refined 2D and 3D AutoCAD Proficiency** We significantly improved our ability to navigate the AutoCAD environment, bridging the gap between flat drafting and volumetric modeling. This project taught us to seamlessly integrate 2D drafts with 3D commands, allowing us to visualize how a simple cross-section transforms into a complex solid using tools like Extrude, Revolve, and Loft.
- **Understanding of Technical Terminologies & Concepts** We moved beyond basic tool usage to a logical understanding of core CAD concepts. We gained a strong command over fundamental terminologies such as Boolean operations (Union, Subtract). This ensured we could not only execute commands but strictly understand the underlying geometry the software was creating.
- **Strengthened Precision and Assembly Logic** Designing a functional engine required absolute accuracy. We learned that "close enough" results in mechanical failure, forcing us to adopt a disciplined approach to dimensional precision. We very precisely align multiple moving parts ensuring the pistons, rods, and crankshaft mated perfectly without interference or collision.
- **Mastery of the User Coordinate System (UCS)** One of the most critical technical takeaways was mastering the UCS. Working on the inclined banks of a V-engine forced us to manipulate the X, Y, and Z axes dynamically.
- **Practical Problem-Solving through Hands-On Application** This project provided deep learning that theoretical study cannot offer. By facing real-world obstacles such as software limitations regarding shelling or the need to assume missing dimensions we learned to improvise and adapt. This hands-on practice transformed our abstract knowledge into practical, execute-ready engineering skill

6. References

- Engine Block
- Cylinder Head
- Valve Cover
- Rocker Arm
- Rocker Arm Assembly
- Piston Complete Assembly
- Side Box
- Oil Pan
- Belt Wheel 1
- Belt wheel 2
- Bushes
- Camshaft
- Crankshaft
- Camshaft Retainer
- Rocker arm spring
- Air Filter
- Valve
- Intake Manifold
- Exhaust Manifold
- Assembly of engine

7. BOM



BOM			
Sr.#	**Qunatity**	**Name**	**Material**
1	x1	Oil Pan	Aluminum
2	x1	Crank Shaft	Cast Iron
3	x1	Engine Head	Cast Iron
4	x2	Exhaust Manifold	Cast Iron
5	x6	Piston	Cast Iron
6	x1	Belt Wheel 1	Steel
7	x1	Engine Side Cover	Steel
8	x1	Crank Shaft Bushing	Aluminum Alloy
9	x2	Engine Belt Wheel 2	Steel
10	x14	M8 Bolt	Steel
11	x1	Cam Shaft	Cast Iron
12	x1	Intake Manifold	Cast Iron
13	x2	Valve Cover	Aluminum Alloy
14	x2	Cylinder Head	Cast Iron
15	x8	Camshaft Retainer	Aluminum Aloy
16	x12	Engine Valve	Cast Iron
17	x12	Rocker Arm	Cast Iron
18	x12	Bolt	Steel(ISO 8676)
19	x12	Rocker Spring	Chromium -vanadium Steel Alloy
20	x2	Air Filter	Carbon Steel

Name : Muaaz Ahmad	Roll : 2502338
Name : Tayyab Mahmood Adil	Roll : 2502272
Project : V-Line Engine	Class: BEME-F25-B
Exploded Assembly Of Engine	
Scale : 1:4	
All Dimensions in mm	

8. Assembly Tree